

SingleRAN

Common Clock Feature Parameter Description

Issue	Draft B
Date	2026-01-30



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....	1
2 About This Document.....	2
2.1 General Statements.....	2
2.2 Applicable RAT.....	2
2.3 Features in This Document.....	3
3 Common Clock for Multimode Base Stations.....	4
3.1 Principles.....	4
3.1.1 Overview.....	4
3.1.2 Deployment Mode.....	4
3.1.3 Shared Clock Sources.....	6
3.1.3.1 Common GPS/BeiDou/Galileo/GLONASS Clock.....	7
3.1.3.1.1 Common GPS Clock in Separate-MPT Base Stations.....	7
3.1.3.1.2 Common GPS Clock in Co-MPT Multimode Base Stations.....	9
3.1.3.1.3 Common GPS Clock in Hybrid-MPT Base Stations.....	9
3.1.3.2 Common RGPS Clock.....	10
3.1.3.2.1 Common RGPS Clock in Separate-MPT Multimode Base Stations.....	11
3.1.3.2.2 Common RGPS Clock in Co-MPT Multimode Base Stations.....	15
3.1.3.2.3 Common RGPS Clock in Hybrid-MPT Base Stations.....	15
3.1.3.3 Common BITS Clock.....	16
3.1.3.4 Common E1/T1 Clock.....	16
3.1.3.4.1 Common E1/T1 Clock in Separate-MPT Multimode Base Stations.....	17
3.1.3.4.2 Common E1/T1 Clock in Co-MPT Multimode Base Stations.....	19
3.1.3.4.3 Common E1/T1 Clock in Hybrid-MPT Multimode Base Stations.....	19
3.1.3.5 Common 1588 V2 Clock.....	20
3.1.3.5.1 Common IEEE1588 V2 Clock in Separate-MPT Multimode Base Stations.....	22
3.1.3.5.2 Common IEEE1588 V2 Clock in Co-MPT Multimode Base Stations.....	25
3.1.3.5.3 Common IEEE1588 V2 Clock in Hybrid-MPT Multimode Base Stations.....	25
3.1.3.6 Common Synchronous Ethernet Clock.....	26
3.1.3.6.1 Common Synchronous Ethernet Clock in Separate-MPT Multimode Base Stations.....	27
3.1.3.6.2 Common Synchronous Ethernet Clock in Co-MPT Multimode Base Stations.....	29
3.1.3.6.3 Common Synchronous Ethernet Clock in Hybrid-MPT Multimode Base Stations.....	30
3.1.3.7 Other Common Reference Clocks.....	31

3.1.3.7.1 Common 1PPS+TOD Clock in Co-MPT Multimode Base Stations.....	31
3.1.3.7.2 Common 1PPS+TOD Clock in Separate-MPT Multimode Base Stations.....	33
3.1.3.7.3 Common IEEE1588 V2+SyncE Clock in Co-MPT Multimode Base Stations.....	33
3.1.3.7.4 Common GNSS+SyncE Clock in Co-MPT Multimode Base Stations.....	36
3.1.3.7.5 Common IEEE1588 V2+SyncE Clock in Separate-MPT Multimode Base Stations.....	37
3.1.3.7.6 Common GNSS+IEEE1588 V2&SyncE Clock in Co-MPT Multimode Base Stations.....	41
3.1.3.7.7 Common GNSS+IEEE1588 V2&SyncE Clock in Separate-MPT Multimode Base Stations.....	44
3.2 Network Analysis.....	46
3.2.1 Benefits.....	46
3.2.2 Impacts.....	46
3.3 Requirements.....	46
3.3.1 Licenses.....	46
3.3.2 Functions.....	46
3.3.2.1 Prerequisite Functions.....	47
3.3.2.2 Mutually Exclusive Functions.....	51
3.3.2.3 Function Impacts.....	51
3.3.3 Hardware.....	52
3.3.4 Others.....	52
3.4 Operation and Maintenance.....	53
3.4.1 When to Use.....	53
3.4.2 Precautions.....	53
3.4.3 Data Configuration.....	54
3.4.3.1 Side Providing the Clock Source.....	54
3.4.3.1.1 eGBTS/NodeB/eNodeB/gNodeB/Co-MPT Base Station Providing the Clock Source.....	54
3.4.3.1.2 GBTS Providing the Clock Source.....	60
3.4.3.2 Sharing Clock Source Provided by Another RAT.....	62
3.4.3.2.1 eGBTS/NodeB/eNodeB/gNodeB/Co-MPT Base Station Sharing Clock Source Provided by Another RAT.....	62
3.4.3.2.2 GBTS Sharing Clock Source Provided by Another RAT.....	63
3.4.4 Activation Verification.....	64
3.4.5 Network Monitoring.....	65
4 Inter-RAT Clock Lock.....	66
4.1 Principles.....	66
4.1.1 Overview.....	66
4.1.2 Clock Lock Conditions and Mode Changes.....	67
4.1.3 Major Standard Selection in Clock Lock.....	69
4.1.3.1 System Clock Lock of a Separate-MPT Base Station Including GBTS.....	69
4.1.3.2 System Clock Lock of a Separate-MPT Base Station Excluding GBTS.....	73
4.1.3.3 Forcible Major Standard for Clock Lock.....	73
4.1.4 Clock Unlock in eCPRI Scenarios.....	74
4.2 Network Analysis.....	76
4.2.1 Benefits.....	77

4.2.2 Impacts.....	77
4.3 Requirements.....	77
4.3.1 Licenses.....	77
4.3.2 Functions.....	77
4.3.2.1 Prerequisite Functions.....	77
4.3.2.2 Mutually Exclusive Functions.....	77
4.3.2.3 Function Impacts.....	77
4.3.3 Hardware.....	77
4.3.4 Others.....	78
4.4 Operation and Maintenance.....	78
4.4.1 When to Use.....	78
4.4.2 Precautions.....	78
4.4.3 Data Configuration.....	79
4.4.3.1 Data Preparation.....	79
4.4.3.2 Using MML Commands.....	80
4.4.4 Activation Verification.....	81
4.4.5 Network Monitoring.....	82
5 Parameters.....	84
6 Counters.....	86
7 Glossary.....	87
8 Reference Documents.....	88

1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Revised descriptions in this document.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 02 (2025-08-30).

- Technical changes
None
- Editorial changes
Added a restriction that applies to Common Clock in separate-MPT scenarios.
For details, see [3.1.2 Deployment Mode](#).
Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR. This feature works the same way with each of these RATs.

This document describes Common Clock, which applies when BBU sharing is implemented on a multimode base station. BBU sharing means that multiple RATs

such as GSM, UMTS, LTE, and NR are deployed on one logical BBU, which can be a single BBU or two interconnected BBUs.

BBU refers to BBU3900/BBU5900/BBU5900A/BBU3910/BBU5910/BBU3910A/BBU3910C. For details, see *Hardware Description* of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

The BBU3910A/BBU3910C/BBU5900A does not support separate-MPT or BBU interconnection. In a co-MPT base station, unless otherwise specified, the descriptions and examples of the UMPT also apply to the UMDU/UMDC.

For details about clock configuration and supported clock sources for a single RAT, see *Synchronization* for GSM BSS, WCDMA RAN, eRAN, or 5G RAN.

NOTE

- In this document, GBTS/eGBTS, NodeB, eNodeB, and gNodeB correspond to the GSM, UMTS, LTE, and NR modes of a multimode base station, respectively.
- Unless otherwise specified, in this document, LTE and eNodeB always include FDD, TDD, and NB-IoT.
- Unless otherwise specified, in this document, NR and gNodeB always include FDD and TDD.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

RAT	Feature ID	Feature Name	Chapter/Section
GSM	MRFD-121 117	Multi-mode BS Common Reference Clock(GBTS)	3 Common Clock for Multimode Base Stations
UMTS	MRFD-121 127	Multi-mode BS Common Reference Clock(NodeB)	
LTE FDD	MRFD-121 137	Multi-mode BS Common Reference Clock(eNodeB)	
LTE TDD	MRFD-121 147	Multi-mode BS Common Reference Clock(LTE TDD)	
NB- IoT	MRFD-121 157	Multi-mode BS Common Reference Clock(NB-IoT)	
NR	MRFD-151 166	Multi-mode BS Common Reference Clock(NR)	

3 Common Clock for Multimode Base Stations

3.1 Principles

3.1.1 Overview

In Common Clock, all RATs of a multimode base station share a clock source. In common clock mode, a multimode base station requires only one clock source, and all RATs synchronize with this clock source. This reduces the number of clock sources to be deployed and helps cut costs.

The following clock sources can be shared in multimode base stations: GPS/BeiDou/Galileo/GLONASS clock, BITS clock, E1/T1 clock, IEEE 1588V2 clock, synchronous Ethernet (SyncE) clock, 1PPS+TOD clock, IEEE 1588V2+SyncE clock, and GNSS+SyncE clock. The IEEE 1588V2 clock and SyncE clock are feature enhancements. Each RAT has separate license control items for these two types of clocks. When Common Clock is activated, the license of IEEE 1588V2 clock or SyncE clock for a single RAT can be shared by other RATs.

3.1.2 Deployment Mode

Common Clock applies to the following scenarios, depending on the board deployment mode:

Separate-MPT Scenarios

For a separate-MPT multimode base station, each RAT can be connected to an external clock source or use the clock source provided by another RAT. To reduce configuration conflicts, save clock sources, and simplify configurations, it is recommended that a multimode base station use the common clock. The main control board of a RAT may provide a clock source to be shared or use a clock source that is shared by the main control board of another RAT. If all RATs in a multimode base station are configured in the same BBU, common clock is implemented with clock signals transmitted through the BBU backplane. If the

RATs of a multimode base station are configured in two BBUs, common clock is implemented with clock signals transmitted through BBU interconnection cables.

GSM, UMTS, LTE, and NR can provide a clock source to other RATs or use a clock source provided by another RAT. The synchronization mode is configured and controlled on the main control board of the local RAT, regardless of whether the RAT provides the clock source or not. For example, if the LTE mode requires frequency synchronization, the LTE main control board must be configured with frequency synchronization. For a newly deployed separate-MPT multimode base station, the common reference clock is introduced to the main control board of each RAT in descending order of priority as follows: UMPT > GTMU. If all the RATs use the UMPT boards, the common reference clock is introduced to one of the UMPT boards. When common clock reconstruction is involved, the main control board to which the common reference clock is introduced is selected based on the principle of minimum impact on services on the live network.

The following restrictions apply to Common Clock in separate-MPT scenarios:

- If frequency synchronization or hybrid synchronization is configured for the RAT providing a clock source to be shared, only frequency synchronization can be configured for the RATs using the shared clock source.
- If time synchronization is configured for the RAT providing a clock source to be shared, either time synchronization or frequency synchronization can be configured for the RATs using the shared clock source. If time synchronization is configured, services must meet the requirements of loose time synchronization.
- If loose time synchronization is configured for the RAT providing a clock source to be shared, only loose time synchronization or frequency synchronization can be configured for the RATs using the shared clock source.
- If IEEE 1588V2 hybrid synchronization in time and frequency backup mode, or hybrid synchronization in GNSS time and IEEE 1588V2 frequency backup mode is configured for the RAT providing a clock source to be shared, only the frequency can be synchronized to the RATs using the shared clock source. Hybrid synchronization cannot be configured for the RATs using the shared clock source.
- If low-precision frame synchronization is configured for the RAT providing a clock source to be shared, only frequency synchronization can be configured for the RATs using the shared clock source.
- If a multimode base station uses the USCU board for clock synchronization, other clock sources cannot be configured. The main control boards of various RATs use the shared USCU clock for synchronization.
- On a multimode base station, if a RAT is configured with a GNSS+SyncE clock source, other RATs can be configured with only the IEEE 1588V2 reference clock. The GNSS+SyncE reference clock cannot be shared.
- On a multimode base station, if two RATs are configured with an IEEE 1588V2+SyncE clock source, the GNSS clock source cannot be configured as a backup clock source, and inter-RAT clock sharing is not supported.
- On a multimode base station, if a RAT is configured with an IEEE 1588V2+SyncE clock source, the GNSS clock source can be configured as a backup clock source. The other RAT cannot be configured with the IEEE 1588V2+SyncE clock source. In addition, the GNSS, SyncE, ToD, and LineCLK clock sources are not recommended. It is recommended that the other RAT

receive clock signals from the RAT configured with the IEEE 1588V2+SyncE clock source.

- In Cloud BB scenarios, the GNSS+SyncE and IEEE 1588V2+SyncE clock sources cannot be configured, and the USCU board cannot be used to configure clock sources.
- If the USCU board is configured in two RATs, the two RATs must be both configured with the GNSS clock source.

NOTE

To achieve BBU interconnection in different scenarios, you can connect one UCIU board to one UMPT board or connect two UMPT boards. The configurations in the two scenarios are the same. This document uses one interconnection scenario as an example. The BBU3910A/BBU3910C does not support BBU interconnection. For details about the RAT deployment supported in BBU interconnection scenarios, see the "Application Scenarios" sections in *BBU Interconnection*.

In inter-subrack interconnection SDR scenarios, it is not permitted to configure frequency synchronization for the RAT in the primary BBU or time synchronization for the RAT in the secondary BBU at the same time.

The clock source PeerCLK at the local end cannot be shared with the peer end. This restriction is to prevent a clock synchronization loop.

The **DSP PHASEDIFF** command can be used to query the offset between the phases of GNSS clock sources, interconnection clock sources, TOD clock sources configured on a USCU, IP clock sources, and peer clock sources. When the peer clock works as the standby clock source, only the UMPTe/UMPTg/UMPTga/UMPTba supports the query of the phase offset.

Co-MPT Scenarios

The clock source configured for the shared main control board is shared by each RAT and Common Clock is supported by default. Both frequency synchronization and time synchronization are supported. If LTE TDD or NR TDD is involved, the clock sources of the RATs sharing a main control board must support and use time synchronization. In this case, GSM, UMTS, and LTE FDD sharing a main control board passively use the time synchronization mode, which affects basic service performance. For details about the impact and workarounds, see *Synchronization*.

Hybrid-MPT Scenarios

A single RAT uses a separate main control board, and the other RATs share a main control board.

- Common Clock is supported by default for the RATs that share the main control board. The clock source supports frequency synchronization and time synchronization. If LTE TDD or NR TDD is involved, the clock source of the co-MPT board must support and use time synchronization.
- The RAT that uses a separate main control board shares a clock source with the RATs that use the shared main control board. The restrictions in this scenario are the same as those in separate-MPT scenarios.

3.1.3 Shared Clock Sources

3.1.3.1 Common GPS/BeiDou/Galileo/GLONASS Clock

GPS, BeiDou, Galileo, and GLONASS are satellite navigation systems. Therefore, a BeiDou/Galileo/GLONASS clock has the same working principles and networking schemes as those of a GPS clock.

A BeiDou/Galileo/GLONASS clock is different from a GPS clock in the configuration and hardware aspects as follows:

- Configuration: The clock working mode must be set to BeiDou/Galileo/GLONASS for a BeiDou/Galileo/GLONASS clock source.
- Hardware: A board and an antenna system that support BeiDou/Galileo/GLONASS clock synchronization are required.

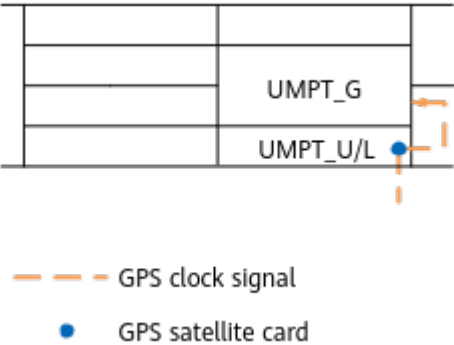
The following sections use the GPS clock as an example to describe the networking schemes for common GPS/BeiDou/Galileo/GLONASS clock in separate-MPT, co-MPT, and hybrid-MPT multimode base stations.

3.1.3.1.1 Common GPS Clock in Separate-MPT Base Stations

Single BBU Scenarios

Both the UMPT and USCU can introduce GPS clock signals. The boards in the BBU use the GPS clock signals through the BBU backplane. The main control board that introduces GPS clock signals must contain a GPS satellite card. If the USCU that introduces GPS clock signals is configured for both RATs, the two RATs deployed on the separate-MPT base station cannot both use time synchronization. One RAT can use time synchronization and provides clock signals to the other RAT. The USCU board occupies an extra slot in the BBU. Therefore, it is recommended that the main control board be connected to an external GPS clock source, as shown in [Figure 3-1](#) and [Figure 3-2](#).

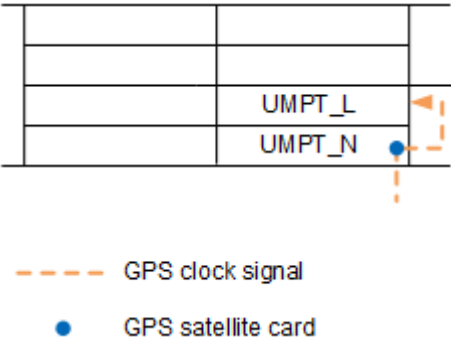
Figure 3-1 Common GPS clock in a GU/GL separate-MPT base station



NOTE

The UMPT_G in the figure can be replaced by the GTMU.

Figure 3-2 Common GPS clock in an LN separate-MPT base station



BBU Interconnection Scenarios

The following describes the principle of clock sharing in BBU interconnection scenarios.

- As shown in Figure 3-3, the GPS clock is located in the primary BBU. The primary and secondary BBUs are interconnected through the CI ports on the panels. The primary BBU provides GPS clock signals for the secondary BBU. The RAT that provides the clock source to be shared is configured with time synchronization.
- As shown in Figure 3-4, the GPS clock is located in the secondary BBU. The secondary BBU sends GPS clock signals to the primary BBU through the CI port on the UMPT board. The boards in the primary BBU use the GPS clock signals through the BBU backplane.

For details about inter-BBU interconnection, see "Application Scenarios" in *BBU Interconnection*. This section describes the process of inter-RAT GPS clock sharing in GULN quadruple-mode inter-BBU scenarios.

Figure 3-3 Common GPS clock in the primary BBU of a GU+LN base station

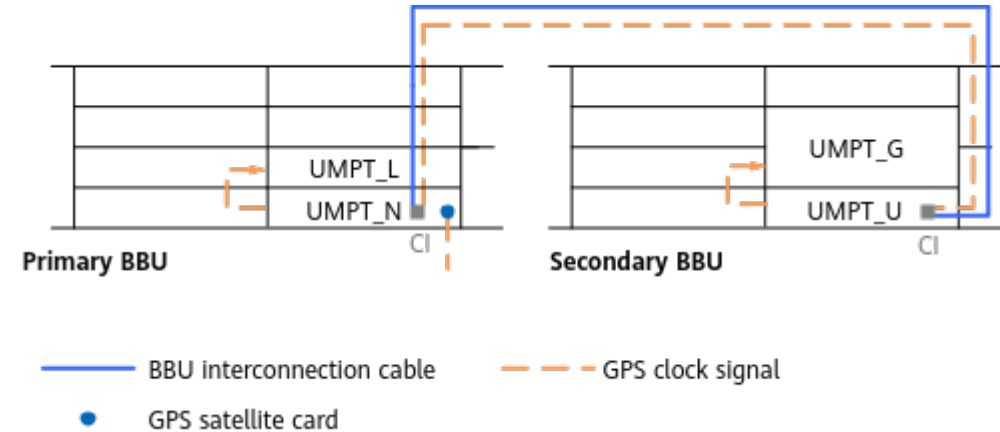
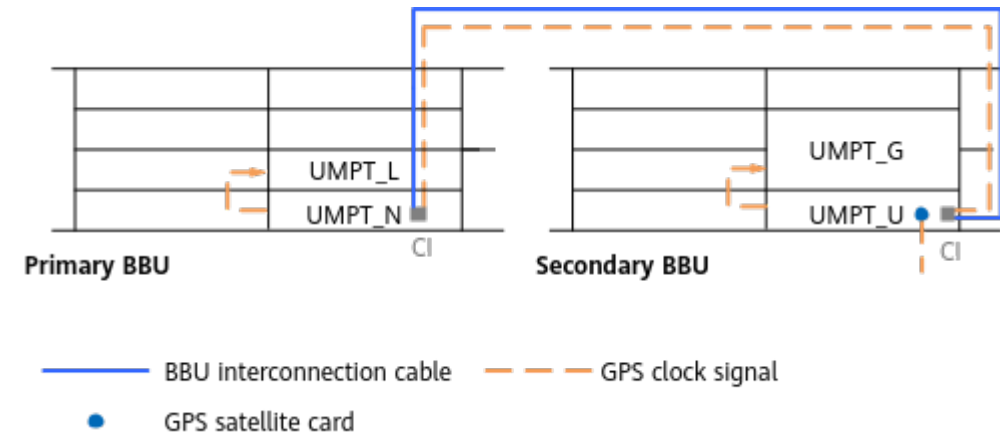


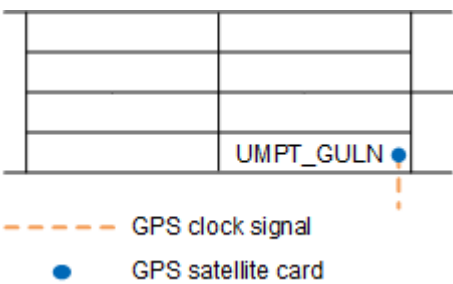
Figure 3-4 Common GPS clock in the secondary BBU of a GU+LN base station



3.1.3.1.2 Common GPS Clock in Co-MPT Multimode Base Stations

As shown in Figure 3-5, RATs in a GULN multimode base station share a main control board and support common GPS clock by default.

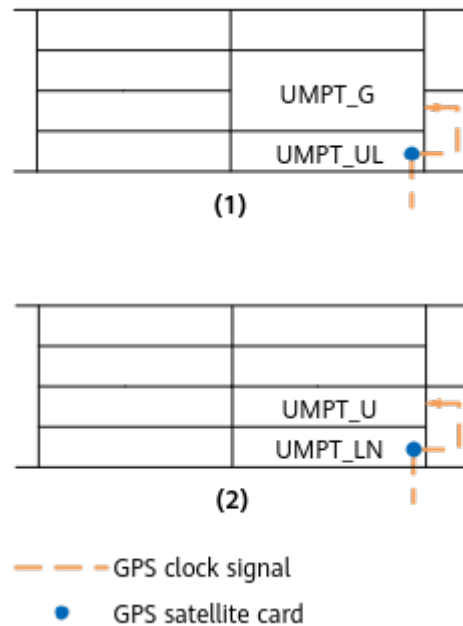
Figure 3-5 Common GPS clock in a GULN co-MPT base station



3.1.3.1.3 Common GPS Clock in Hybrid-MPT Base Stations

Single BBU Scenarios

As shown in Figure 3-6, RATs in a single BBU share the GPS clock through the BBU backplane.

Figure 3-6 Common GPS clock in a hybrid-MPT base station

BBU Interconnection Scenarios

The common clock in BBU interconnection scenarios of hybrid-MPT base stations works in the same way as that in BBU interconnection scenarios of separate-MPT base stations. For details, see [BBU Interconnection Scenarios](#) in [3.1.3.1 Common GPS Clock in Separate-MPT Base Stations](#).

3.1.3.2 Common RGPS Clock

This section describes the common RGPS clock for a separate-MPT or co-MPT multimode base station. The RGPS clock signals can be obtained when the RF modules are connected to an external RGPS reference source. NR in this section refers to TDD.

In this scenario, the common RGPS reference clock has the following restrictions:

- The RGPS clock must be configured for the major standard for clock lock. For example, in inter-BBU scenarios, the RGPS clock must be configured in the primary BBU.
- All NR-capable AAUs with AUX ports for RGPS support the common RGPS clock function. For details, see the hardware description of the related AAUs in *3900 & 5900 Series Base Station Product Documentation*.
- When multimode RF modules are used, RGPS cannot be configured for multiple modes at the same time.
- The external RGPS clock source of RF modules and the external GPS clock source of the BBU cannot be configured at the same time.
- If the RGPS clock needs to be shared with other modes, the base station must be upgraded to SRAN17.0 or later.

3.1.3.2.1 Common RGPS Clock in Separate-MPT Multimode Base Stations

Single BBU Scenarios

In LN separate-MPT scenarios where a single BBU is configured and the common RGPS clock is used, the BBU must be a BBU5900/BBU5910/BBU3910. The BBU5900 is used as an example. The following scenarios are supported:

- As shown in [Figure 3-7](#), RF modules on the NR side are connected to the RGPS clock source. The NR main control board obtains clock signals and shares them with the LTE main control board.
- As shown in [Figure 3-8](#), when multimode RF modules are used, clock lock is used between LTE and NR. In the following example, the RF modules on the NR side are connected to the RGPS reference source. In this case, NR must be the major standard for clock lock, and the LTE side cannot be connected to an external reference source such as CGPS.
- As shown in [Figure 3-9](#), LTE and NR share RF modules and use an external RGPS reference clock, and clock lock is used between LTE and NR. In this case, one mode serves as the major standard for clock lock, and the other mode cannot be connected to a reference source such as CGPS.

Figure 3-7 Common RGPS clock in an LN separate-MPT base station when the NR side receives RGPS clock signals

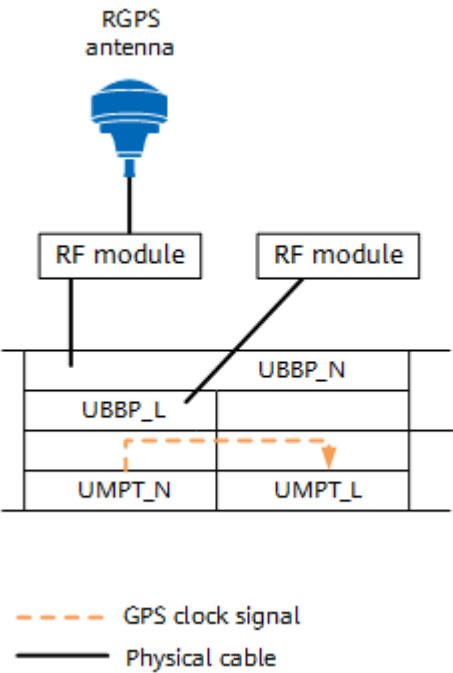


Figure 3-8 Clock lock in an LN separate-MPT base station when the NR side receives RGPS clock signals

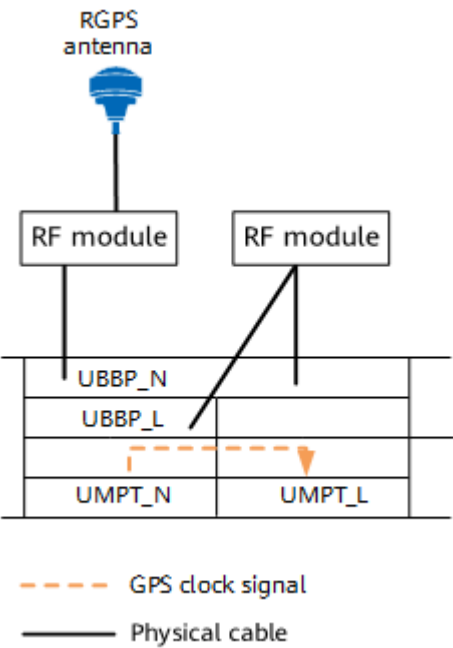
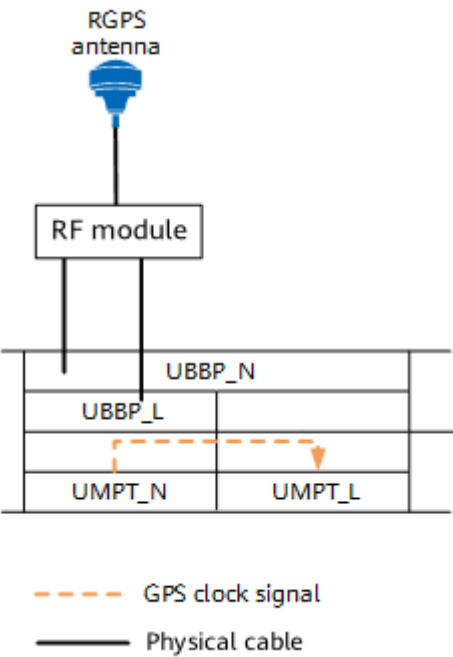


Figure 3-9 Clock lock in an LN separate-MPT base station when a multimode RF module is used and the NR side receives RGPS clock signals



BBU Interconnection Scenarios

In separate-MPT scenarios where BBUs are interconnected and the RGPS clock is shared, LTE and NR must be deployed in different BBUs, and CI interconnection is used between BBUs. The BBU on the NR side must be a BBU5900/BBU5910/BBU3910. When clock lock is used, the RGPS clock source is deployed in the

- primary BBU. The BBU5900 is used as an example. The following scenarios are supported:
- As shown in [Figure 3-10](#), RF modules on the NR side are connected to an external RGPS clock source and share clock signals with LTE in the secondary BBU.
 - As shown in [Figure 3-11](#), when multimode RF modules are used, clock lock is used between LTE and NR. In the following example, the RF modules on the NR side are connected to the RGPS reference source. In this case, NR must be the major standard for clock lock, and the LTE side cannot be connected to an external reference source such as CGPS.
 - As shown in [Figure 3-12](#), LTE and NR share RF modules and use an external RGPS reference clock, and clock lock is used between LTE and NR. In this case, one mode serves as the major standard for clock lock, and the other mode cannot be connected to a reference source such as CGPS.

Figure 3-10 Common RGPS clock in an LN separate-MPT base station with BBU interconnection

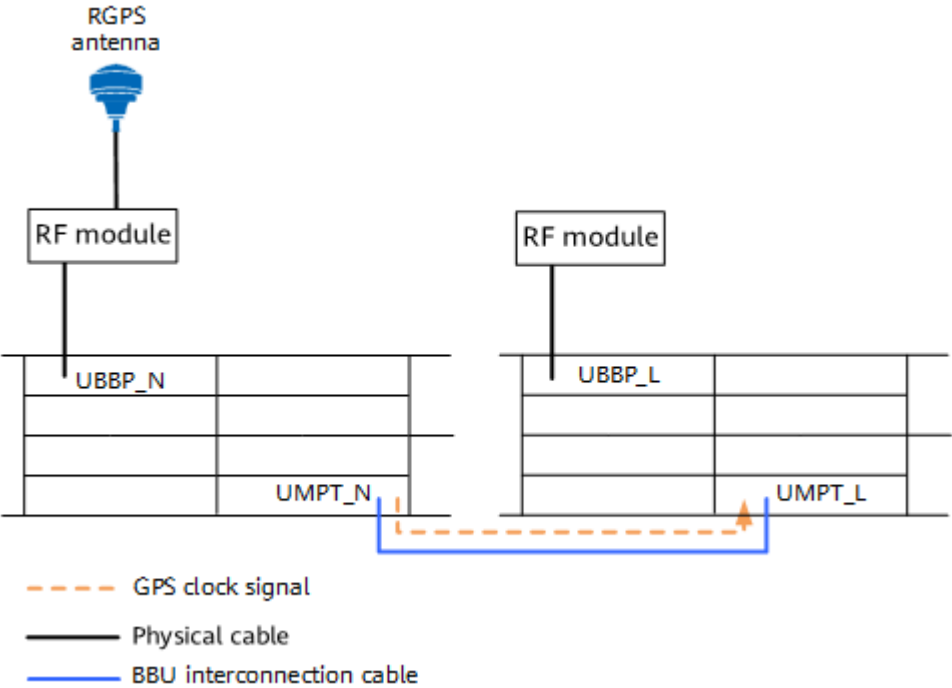


Figure 3-11 RF RGPS clock on the NR side of clock lock in an LN separate-MPT base station with BBU interconnection

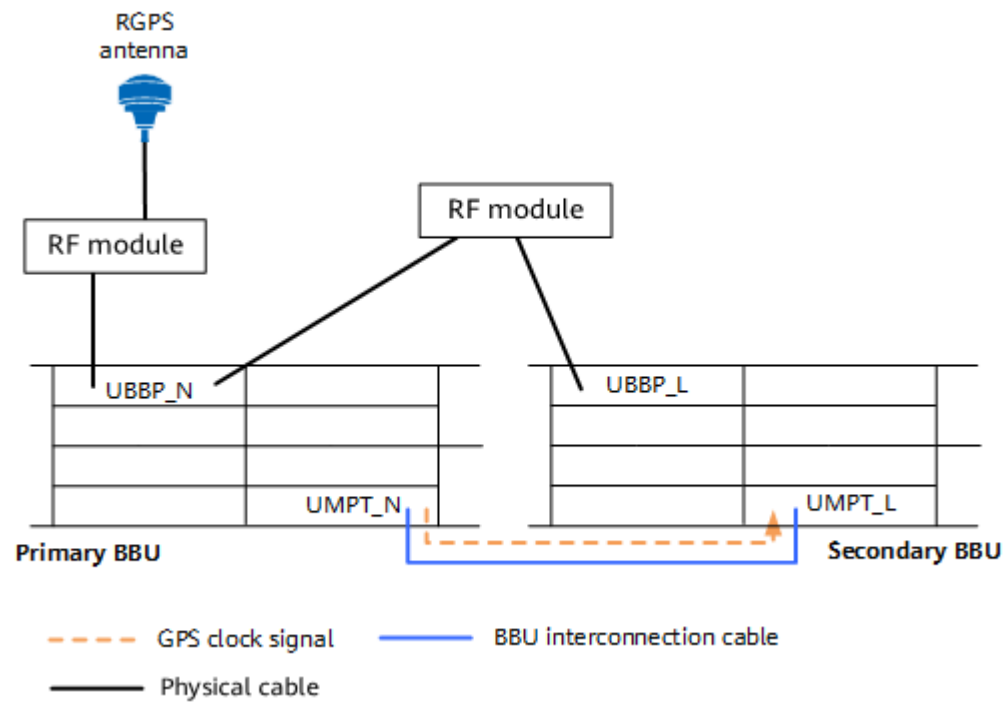
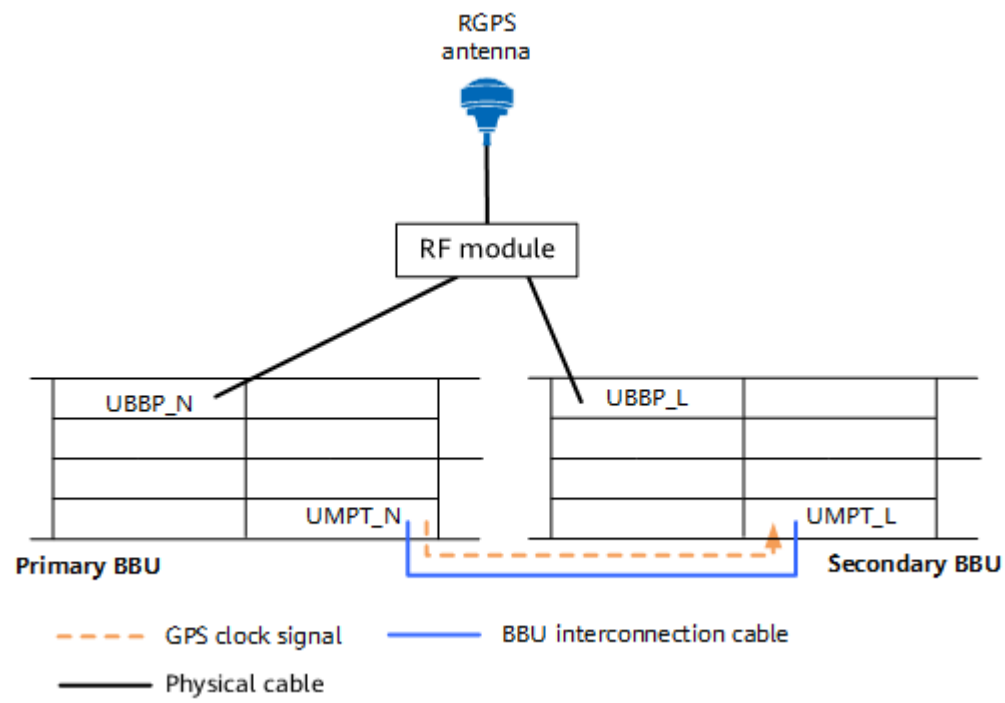


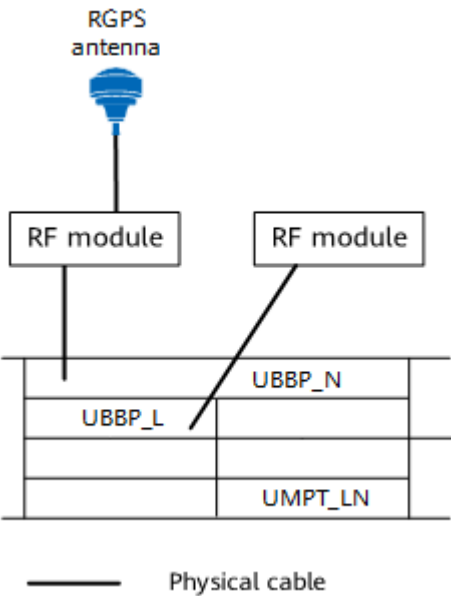
Figure 3-12 Co-RF RGPS clock on the NR side of clock lock in an LN separate-MPT base station with BBU interconnection



3.1.3.2.2 Common RGPS Clock in Co-MPT Multimode Base Stations

In an LN co-MPT base station using the common RGPS clock, the BBU can be a BBU5900/BBU5910/BBU3910/BBU5900A. The BBU5900 is used as an example, as shown in [Figure 3-13](#).

Figure 3-13 Common RGPS clock in an LN co-MPT base station

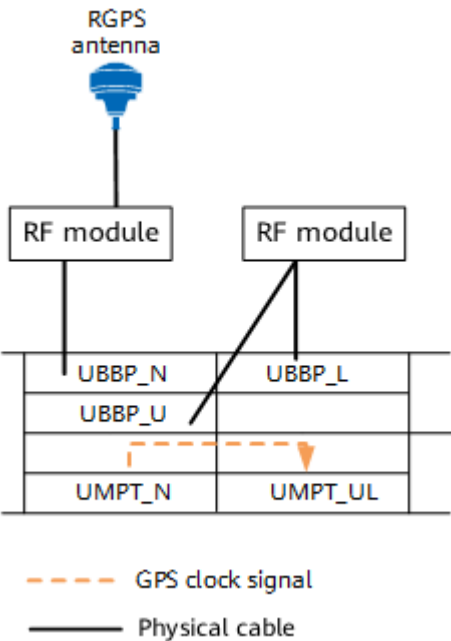


3.1.3.2.3 Common RGPS Clock in Hybrid-MPT Base Stations

Single BBU Scenarios

In a base station where a single BBU is configured and the common RGPS clock is used, the BBU5900/BBU5910/BBU3910 must be used. The BBU5900 is used as an example, as shown in [Figure 3-14](#).

Figure 3-14 Common RGPS clock in a ULN hybrid-MPT multimode base station



BBU Interconnection Scenarios

The common clock in BBU interconnection scenarios of hybrid-MPT base stations works in the same way as that in BBU interconnection scenarios of separate-MPT base stations. For details, see [BBU Interconnection Scenarios](#) in [3.1.3.2.1 Common RGPS Clock in Separate-MPT Multimode Base Stations](#).

3.1.3.3 Common BITS Clock

This section describes the common BITS clock in separate-MPT, co-MPT, and hybrid-MPT multimode base stations. The BITS clock only supports frequency synchronization and does not support time synchronization. In this section, "LTE" refers to both LTE FDD and LTE NB-IoT.

The BITS clock source must be configured on the USCU board, which occupies an extra slot. In addition, the BITS clock source cannot be transmitted over the transport network, which increases costs. Therefore, the BITS clock source is not recommended.

The sharing modes of the BITS clock source are similar to those of the GNSS clock source.

3.1.3.4 Common E1/T1 Clock

This section describes the common E1/T1 clock in separate-MPT, co-MPT, and hybrid-MPT multimode base stations. The E1/T1 clock only supports frequency synchronization and does not support time synchronization. NR does not support the E1/T1 clock. In this section, "LTE" refers to both LTE FDD and LTE NB-IoT, and "eNodeB" refers to both FDD eNodeB and NB-IoT eNodeB.

[Table 3-1](#) describes the typical application scenarios for common E1/T1 clock.

Table 3-1 Typical application scenarios for common E1/T1 clock

Common Clock Solution	Description
Common E1/T1 clock over the Abis interface	When the Abis interface uses E1/T1 transmission and the lub/S1 interface uses IP over FE/GE transmission, the NodeB/eNodeB can obtain E1/T1 clock signals from an E1/T1 link over the Abis interface.
Common E1/T1 clock over the lub interface	When the Abis interface on the multimode base station side uses IP co-transmission or IP hybrid transmission, the eGBTS/GBTS can obtain E1/T1 clock signals from an E1/T1 link over the lub interface.
	When the lub interface uses E1/T1 transmission and the S1 interface uses IP over FE/GE transmission, the eNodeB can obtain E1/T1 clock signals from an E1/T1 link over the lub interface.

NOTE

- Only UMTS can obtain E1/T1 clock signals from a UTRP.
- LTE can obtain E1/T1 clock signals only from a UMPT.

For details about the Common Transmission feature, see *Common Transmission*.

3.1.3.4.1 Common E1/T1 Clock in Separate-MPT Multimode Base Stations

Single BBU Scenarios

As shown in Figure 3-15 and Figure 3-16, one RAT of a multimode base station extracts E1/T1 clock signals from an E1/T1 link and provides the clock signals to the other RATs. The other RATs obtain the clock signals through the BBU backplane.

Figure 3-15 Common E1/T1 clock in a GU/GL separate-MPT base station

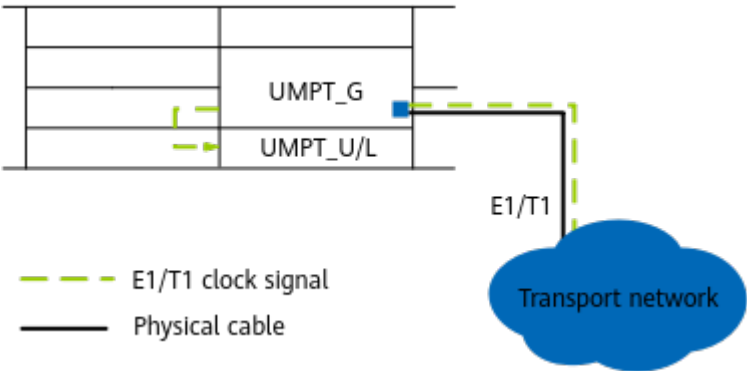
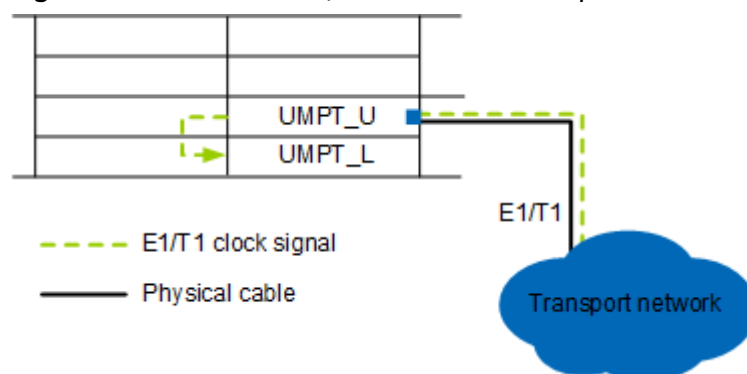


Figure 3-16 Common E1/T1 clock in a UL separate-MPT base station



BBU Interconnection Scenarios

The following uses the BBU interconnection achieved by connecting one UCIU to one UMPT as an example. One RAT is configured with an E1/T1 clock source and shares it with the other two RATs. Clock sharing works as follows:

- As shown in [Figure 3-17](#), the E1/T1 clock source is located in the primary BBU. One RAT in the primary BBU extracts E1/T1 clock signals from an E1/T1 link. The other RAT in the primary BBU synchronizes with the E1/T1 clock source through the BBU backplane. In addition, the clock signals are sent to the secondary BBU through the CI port.
- As shown in [Figure 3-18](#), the E1/T1 clock source is located in the secondary BBU. The RAT in the secondary BBU extracts E1/T1 clock signals from an E1/T1 link. The E1/T1 clock signals are provided to the primary BBU through the CI port on the UMPT. The RATs in the primary BBU use the E1/T1 clock signals through the BBU backplane.

Figure 3-17 Common E1/T1 clock in the primary BBU of a GU+L base station

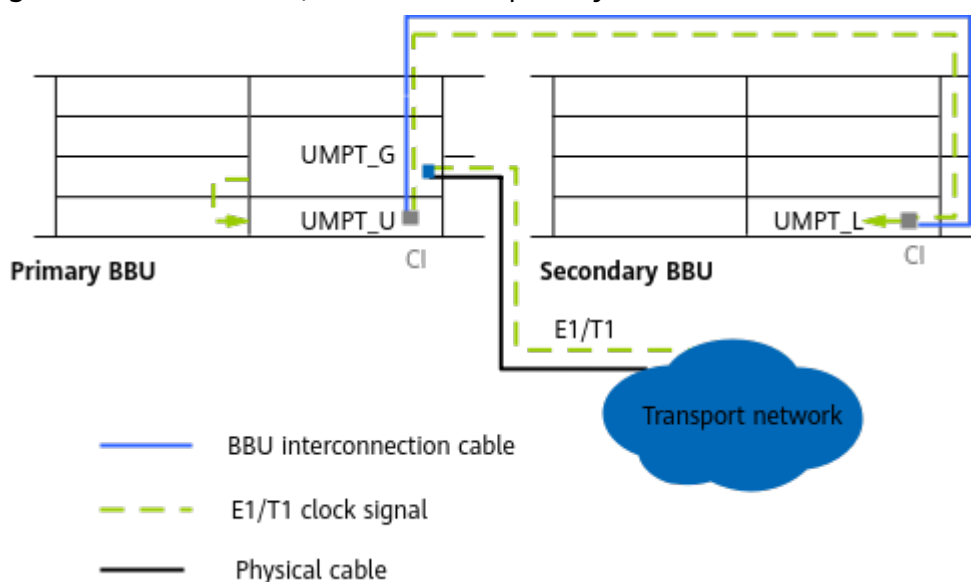
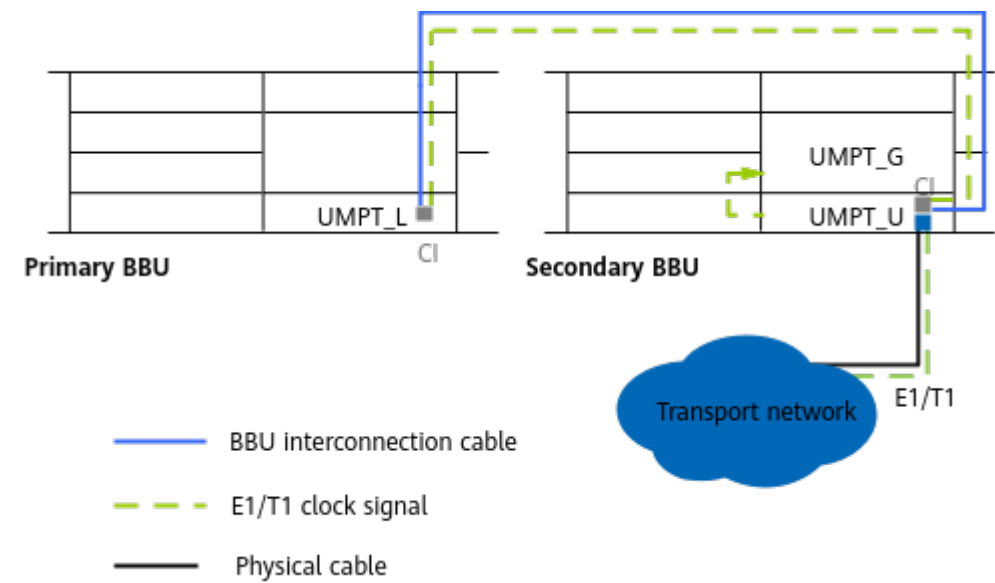


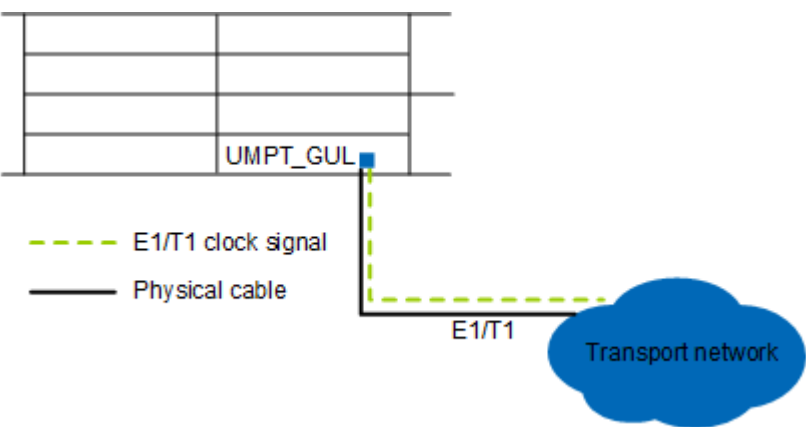
Figure 3-18 Common E1/T1 clock in the secondary BBU of a GU+L base station



3.1.3.4.2 Common E1/T1 Clock in Co-MPT Multimode Base Stations

As shown in Figure 3-19, RATs in a multimode base station share a UMPT, share the E1/T1 clock source configured on the UMPT, and support the Common Clock feature by default.

Figure 3-19 Common E1/T1 clock in a GUL co-MPT base station

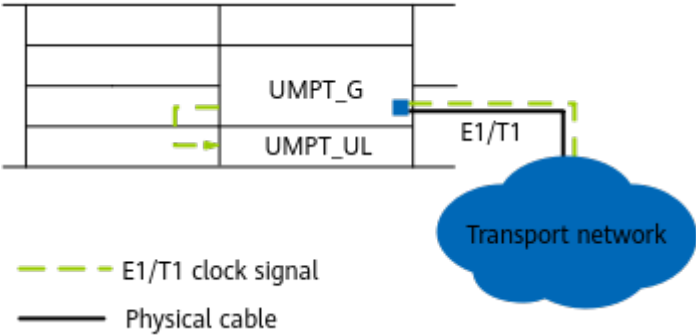


3.1.3.4.3 Common E1/T1 Clock in Hybrid-MPT Multimode Base Stations

Single BBU Scenarios

As shown in Figure 3-20, RATs in a single BBU share the E1/T1 clock source through the BBU backplane.

Figure 3-20 Common E1/T1 clock in a hybrid-MPT multimode base station



BBU Interconnection Scenarios

The common clock in BBU interconnection scenarios of hybrid-MPT base stations works in the same way as that in BBU interconnection scenarios of separate-MPT base stations. For details, see [BBU Interconnection Scenarios](#) in [3.1.3.4.1 Common E1/T1 Clock in Separate-MPT Multimode Base Stations](#).

3.1.3.5 Common 1588 V2 Clock

1588 V2 clocks include IEEE1588 V2, 1588V2 ATR, ITU-T G.8265.1, and ITU-T G.8275.1 clocks, all of which provide IP clock synchronization based on the 1588 V2 protocol. These clocks differ in their synchronization modes and supported RATs, as shown in [Table 3-2](#).

Table 3-2 Synchronization modes and RATs supported by a 1588 V2 clock

Clock Type	Synchronization Mode	Supported RAT
IEEE1588 V2 (Huawei proprietary) L3 unicast	Frequency synchronization and time synchronization	GSM, UMTS, FDD, TDD, NR
IEEE1588 V2 L2 multicast	Time synchronization	GSM, UMTS, FDD, TDD, NR
IEEE1588 V2 16.1 L3 unicast	Frequency synchronization and time synchronization	GSM, UMTS, FDD, TDD, NR
ITU-T G.8275.2 L3 unicast	Time synchronization	GSM, UMTS, FDD, NR FDD
ITU-T G.8265.1 L3 unicast	Frequency synchronization	GSM, UMTS, FDD, NR
ITU-T G.8275.1 L2 multicast	Time synchronization	GSM, UMTS, FDD, TDD, NR

Clock Type	Synchronization Mode	Supported RAT
IEEE1588 V2 (Huawei proprietary) L3 unicast IEEE1588 V2 16.1 L3 unicast ITU-T G.8275.2 L3 unicast	Loose time synchronization ^a	FDD, NR
a: Loose time synchronization is a time synchronization mode with low precision (in milliseconds).		

Table 3-3 describes the typical application scenarios for common 1588 V2 clock.

Table 3-3 Typical application scenarios for common 1588 V2 clock

Common Clock Solution	Description
Common 1588 V2 clock over the lub interface	When the lub interface uses IP transmission and the NodeB is configured with a 1588 V2 clock source over the lub interface, the eGBTS/GBTS can obtain clock signals from the UMPT of the NodeB through the BBU backplane.
Common 1588 V2 clock over the S1 interface	The S1 interface uses IP transmission on the base station side and is configured with 1588 V2 clock synchronization. The eGBTS/GBTS/NodeB/gNodeB can obtain clock signals from the UMPT of the eNodeB through the BBU backplane.
Common IEEE1588 V2 clock over the Abis interface	When the eGBTS or GBTS is configured with a 1588V2 clock source over the Abis interface, the NodeB and eNodeB can obtain clock signals from the UMPT_G/GTMUc of the eGBTS/GBTS through the BBU backplane.
Common 1588 V2 clock over the NG interface	When the NG interface uses IP transmission and the gNodeB is configured with a 1588 V2 clock source over the NG interface, the eGBTS/GBTS/NodeB/eNodeB can obtain clock signals from the UMPT of the gNodeB through the BBU backplane.

If a common 1588 V2 clock is used in co-transmission scenarios, the shared clock source must be configured on the RAT that provides the co-transmission port, and the RAT that does not provide the co-transmission port uses the clock signals provided by the peer RAT.

If the RAT providing a clock source uses loose time synchronization:

- Only LTE FDD and NR support loose time synchronization with the peer end, and the UMPT board must be used.

- UMTS, LTE, and NR can receive shared loose time synchronization signals. The UMPT supports shared loose time synchronization.
- The GTMU board does not support the loose time synchronization mode. When a multimode separate-MPT base station uses the GTMU board and LTE and NR require loose time synchronization, the clock lock optimization switch on the GSM side must be turned on (that is, **BTSOTHPARA.ClkMutualLockOptSw** must be set to **ON**). This prevents GSM from being the major standard in GL clock lock. For details, see [4 Inter-RAT Clock Lock](#).

The following describes the common 1588 V2 clock in separate-MPT, co-MPT, and hybrid-MPT multimode base stations. The IEEE1588 V2 clock is used as an example to describe Common Clock because 1588V2 clocks have the same networking solution even though they differ in function support, RAT support, and data configuration. For details about the networking solution, see *Synchronization* for GSM BSS, WCDMA RAN, eRAN, and 5G RAN.

The 1588 V2 clock used for a co-MPT base station must support all RATs deployed on the co-MPT base station. For example, a co-MPT base station deployed with TDD cannot use the 1588V2 ATR, loose time synchronization, or ITU-T G.8265.1 clock sources.

3.1.3.5.1 Common IEEE1588 V2 Clock in Separate-MPT Multimode Base Stations

When the IEEE1588 V2 clock is used as a frequency synchronization source, LTE TDD or NR TDD can use time synchronization and share the IEEE1588 V2 clock with other RATs but cannot use the IEEE1588 V2 clock provided by other RATs. This is because TDD supports only time synchronization, but the IEEE1588 V2 clock provided by other RATs for TDD can use only frequency synchronization.

When the IEEE1588 V2 clock is used as a time synchronization source:

- GSM, LTE FDD, LTE NB-IoT, and LTE TDD can provide their time synchronization sources for other RATs.
- UMTS can provide the time synchronization source for other RATs if the UMPT is used to obtain the IEEE1588 V2 clock source.

NOTE

Generally, the NodeB does not require time synchronization. UMTS supports the IEEE1588 V2 clock source for the sake of providing the time synchronization source to other RATs.

Single BBU Scenarios

As shown in [Figure 3-21](#) and [Figure 3-22](#), one RAT is connected to the transport network through a fast Ethernet (FE)/Gigabit Ethernet (GE) port. The RAT is configured with an IEEE1588 V2 clock client to obtain clock signals from an IEEE1588 V2 clock server. The local oscillator of the RAT synchronizes with the clock source and transmits the synchronized 1PPS clock signals to the other RAT through the backplane. The 1PPS clock signals serve as the reference clock source for the other RAT. In this scenario, only one IEEE1588 V2 clock client is required.

Figure 3-21 Common IEEE1588 V2 clock in a GU/GL separate-MPT base station

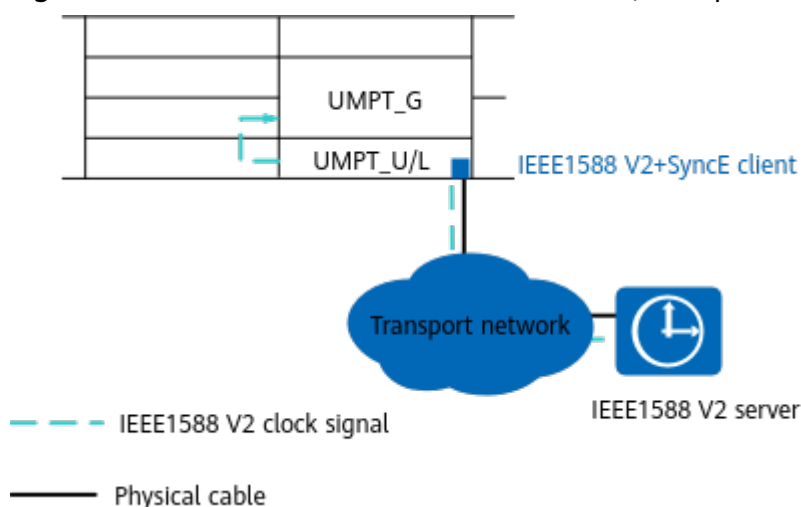
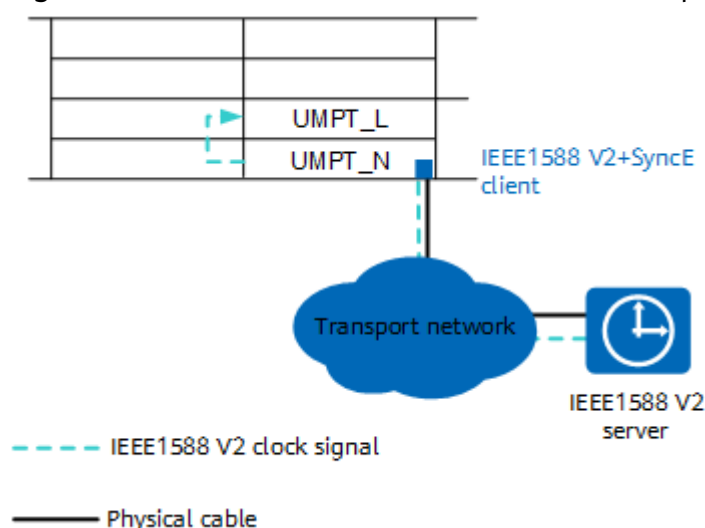


Figure 3-22 Common IEEE1588 V2 clock in an LN separate-MPT base station



BBU Interconnection Scenarios

The following uses the BBU interconnection achieved by connecting UMPTs as an example to describe the principle of clock sharing. One RAT is configured with an IEEE1588 V2 clock source and shares it with the other two RATs.

- As shown in [Figure 3-23](#), the IEEE1588 V2 clock source is located in the primary BBU. One RAT in the primary BBU is configured as an IEEE1588 V2 clock client to receive IEEE1588 V2 clock packets. Then, the RAT generates 1PPS clock signals that are synchronized with the server and forwards them to the secondary BBU through the BBU interconnection port of the UMPT. The other RAT in the primary BBU obtains 1PPS clock signals through the BBU backplane.
- As shown in [Figure 3-24](#), the IEEE1588 V2 clock source is located in the secondary BBU. The RAT in the secondary BBU is configured as an IEEE1588 V2 clock client to receive IEEE1588 V2 clock packets. Then, the RAT generates 1PPS clock signals that are synchronized with the server, and forwards them

to the primary BBU through the BBU interconnection port on the UMPT. The RATs in the primary BBU use the 1PPS clock signals through the BBU backplane.

Transmission port cabling of a multimode base station and the cabling of common IEEE1588 V2 clock are separate.

Figure 3-23 Common IEEE1588 V2 clock in the primary BBU of a GULN base station

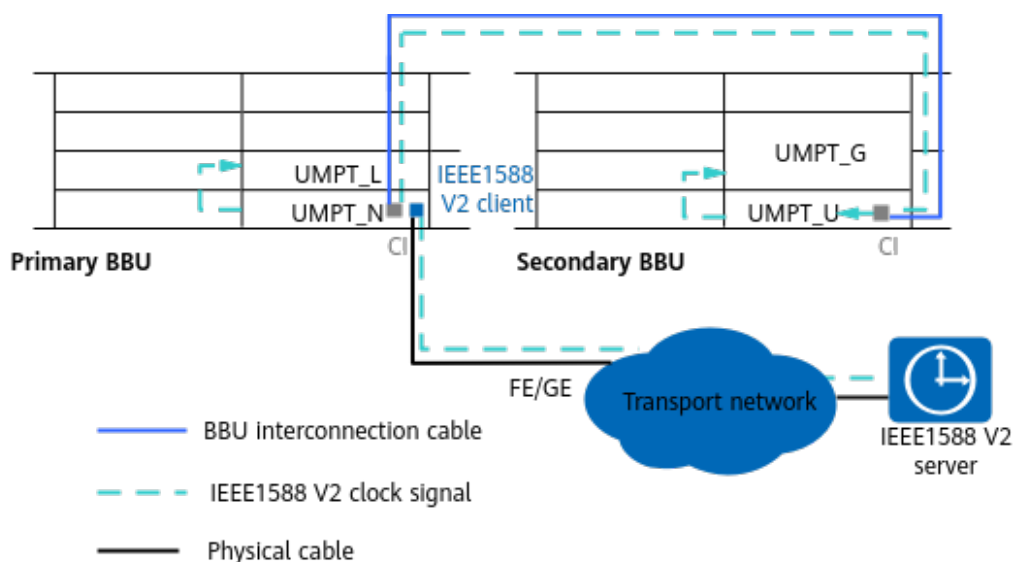
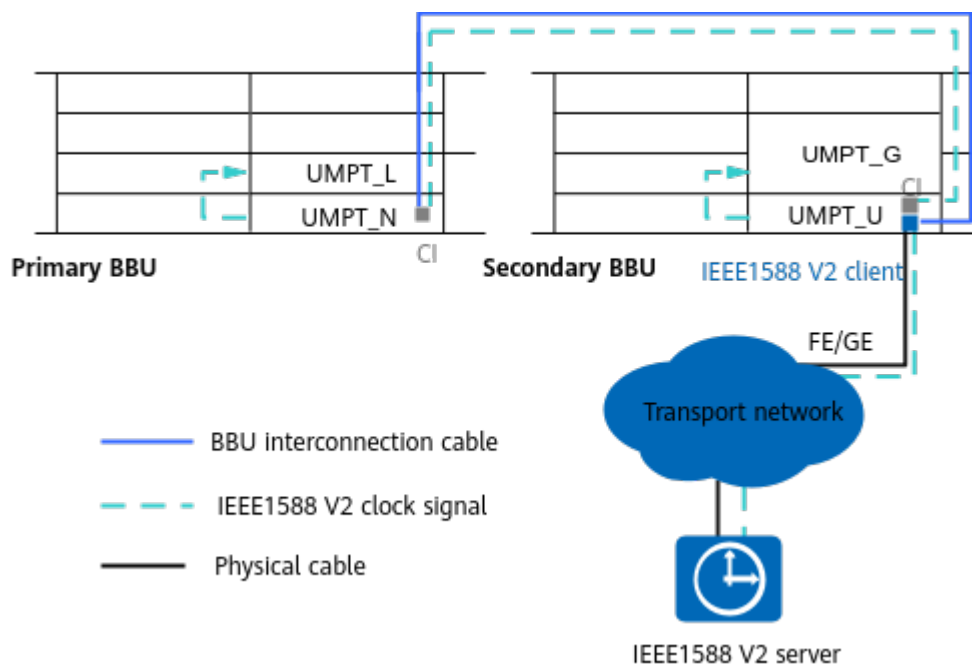


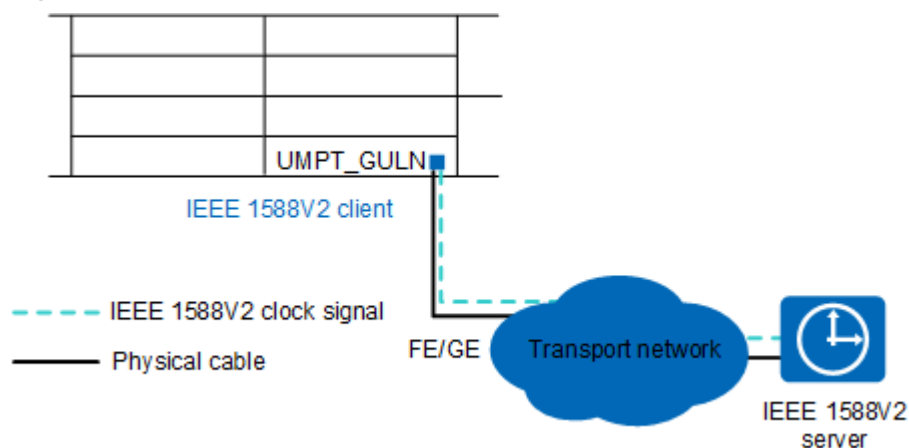
Figure 3-24 Common IEEE1588 V2 clock in the secondary BBU of a GULN base station



3.1.3.5.2 Common IEEE1588 V2 Clock in Co-MPT Multimode Base Stations

As shown in [Figure 3-25](#), RATs in a multimode base station share a UMPT, share the IEEE1588 V2 clock source configured on the UMPT, and support the Common Clock feature by default. The common IEEE1588 V2 clock supports frequency synchronization and time synchronization. If LTE TDD or NR TDD is used, the clock source must use time synchronization.

Figure 3-25 Common IEEE1588 V2 clock in a GULN co-MPT base station

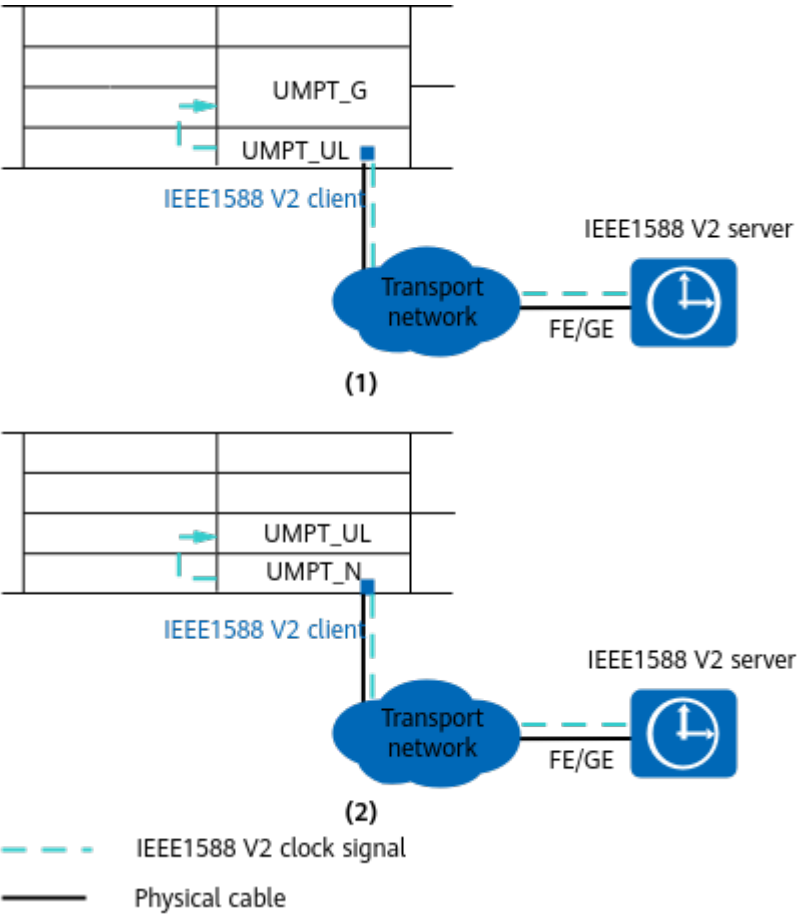


3.1.3.5.3 Common IEEE1588 V2 Clock in Hybrid-MPT Multimode Base Stations

Single BBU Scenarios

As shown in [Figure 3-26](#), only one BBU is used. RATs share an IEEE1588 V2 clock source through the BBU backplane.

Figure 3-26 Common IEEE1588 V2 clock in a hybrid-MPT multimode base station



BBU Interconnection Scenarios

The common clock in BBU interconnection scenarios of hybrid-MPT base stations works in the same way as that in BBU interconnection scenarios of separate-MPT base stations. For details, see [BBU Interconnection Scenarios](#) in [3.1.3.5.1 Common IEEE1588 V2 Clock in Separate-MPT Multimode Base Stations](#).

3.1.3.6 Common Synchronous Ethernet Clock

This section describes the common synchronous Ethernet clock in separate-MPT, co-MPT, and hybrid-MPT multimode base stations. The synchronous Ethernet clock only supports frequency synchronization, and does not support time synchronization. In this section, LTE refers to both LTE FDD and LTE NB-IoT. NR refers to NR FDD.

[Table 3-4](#) describes the typical application scenarios for a common synchronous Ethernet clock.

Table 3-4 Typical application scenarios for a common synchronous Ethernet clock

Deployment Scenario	Description
Common synchronous Ethernet clock over the lub interface	When the lub interface on the multimode base station side uses IP transmission and the IP network supports the synchronous Ethernet clock, the eGBTS/GBTS/eNodeB can obtain clock signals from the NodeB's UMPT, or UTRP (supporting FE/GE transmission) managed by the UMTS mode, through the BBU backplane.
Common synchronous Ethernet clock over the S1 interface	When the S1 interface on the multimode base station side uses IP transmission and the IP network supports the synchronous Ethernet clock, the eGBTS/GBTS/NodeB can obtain clock signals from the eNodeB's UMPT, or UTRP (supporting FE/GE transmission) managed by the LTE mode, through the BBU backplane. The gNodeB can obtain clock signals from the UMPT in the eNodeB through the BBU backplane.
Common synchronous Ethernet clock over the Abis interface	When the eGBTS/GBTS of a multimode base station is configured with a synchronous Ethernet clock source over the Abis interface, the NodeB and eNodeB can obtain clock signals from the eGBTS/GBTS's UMPT/GTMUC, or UTRP (supporting FE/GE transmission) managed by the GSM mode, through the BBU backplane.
Common synchronous Ethernet clock over the NG interface	When the NG interface on the multimode base station side uses IP transmission and the IP network supports the synchronous Ethernet clock, the eGBTS/GBTS/NodeB/eNodeB can obtain clock signals from the gNodeB's UMPT through the BBU backplane.

3.1.3.6.1 Common Synchronous Ethernet Clock in Separate-MPT Multimode Base Stations

Single BBU Scenarios

As shown in [Figure 3-27](#) and [Figure 3-28](#), to implement a common synchronous Ethernet clock, one RAT must connect to the transport network over the FE/GE port and the IP network must support the synchronous Ethernet clock. Then the other RAT receives synchronous Ethernet clock signals through the BBU backplane.

Figure 3-27 Common synchronous Ethernet clock in a GU/GL separate-MPT base station

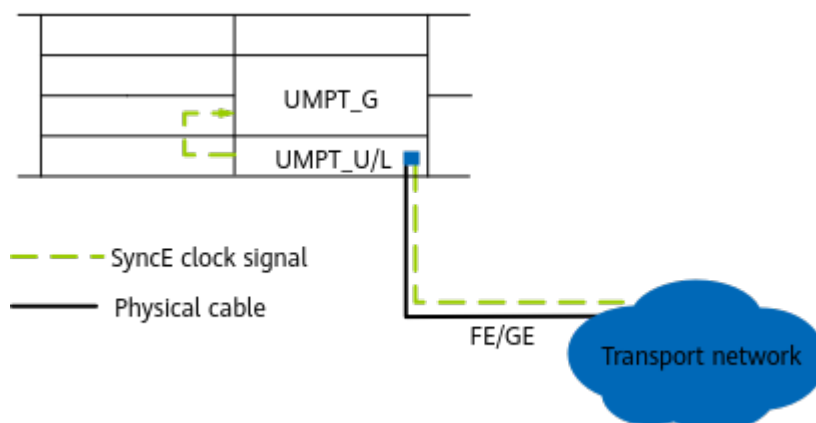
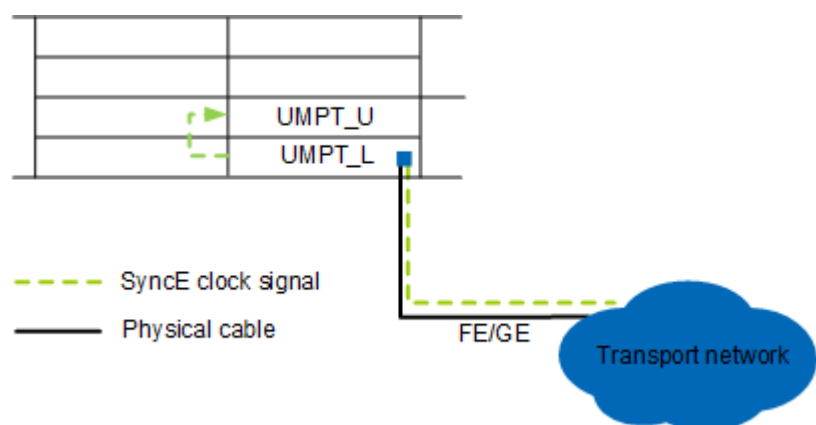


Figure 3-28 Common synchronous Ethernet clock in a UL separate-MPT base station



BBU Interconnection Scenarios

The following uses the BBU interconnection achieved by connecting UMPTs as an example to describe the principle of clock sharing. One RAT is configured with a synchronous Ethernet clock source and provides it to the other two RATs.

- As shown in [Figure 3-29](#), the synchronous Ethernet clock source is located in the primary BBU. One RAT in the primary BBU extracts clock signals from the synchronous Ethernet link and forwards them to the secondary BBU through the UCIU. The other RAT in the primary BBU obtains clock signals through the multimode BBU backplane.
- As shown in [Figure 3-30](#), the synchronous Ethernet clock source is located in the secondary BBU. The RAT in the secondary BBU extracts clock signals from the synchronous Ethernet link, and forwards them to the primary BBU through the BBU interconnection port on the UMPT. The RATs in the primary BBU use the synchronous Ethernet clock signals through the BBU backplane.

Figure 3-29 Common synchronous Ethernet clock in the primary BBU of a GUL base station

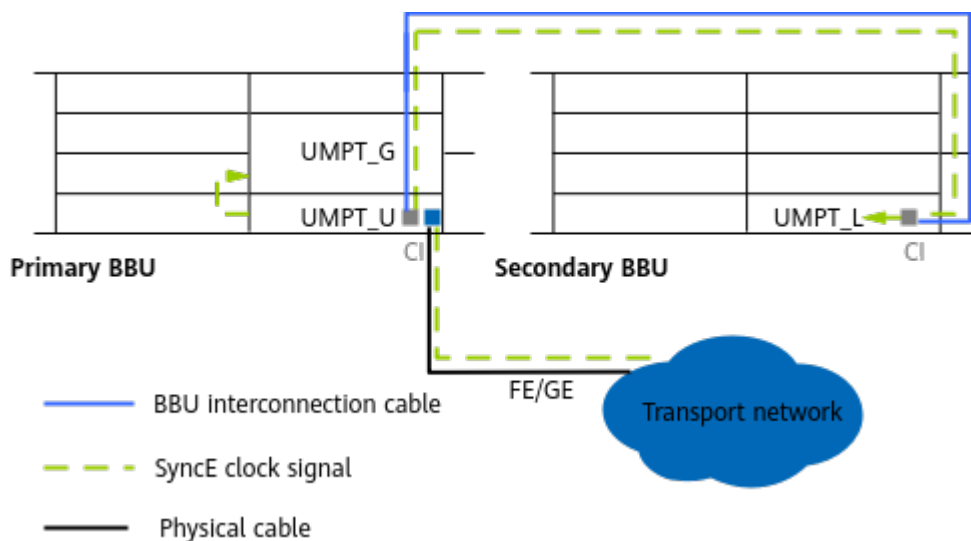
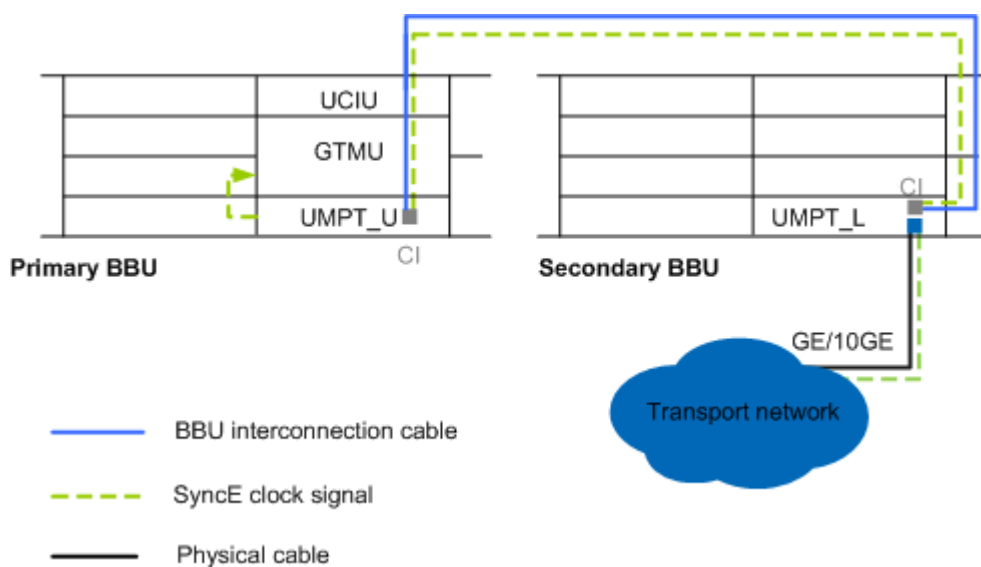


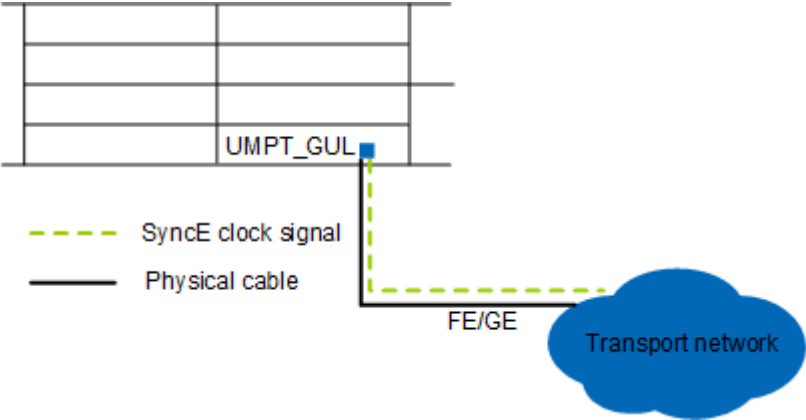
Figure 3-30 Common synchronous Ethernet clock in the secondary BBU of a GUL base station



3.1.3.6.2 Common Synchronous Ethernet Clock in Co-MPT Multimode Base Stations

As shown in [Figure 3-31](#), RATs in a multimode base station share a UMPT, share the synchronous Ethernet clock source configured on the UMPT, and support the Common Clock feature by default.

Figure 3-31 Common synchronous Ethernet clock in a GUL co-MPT base station

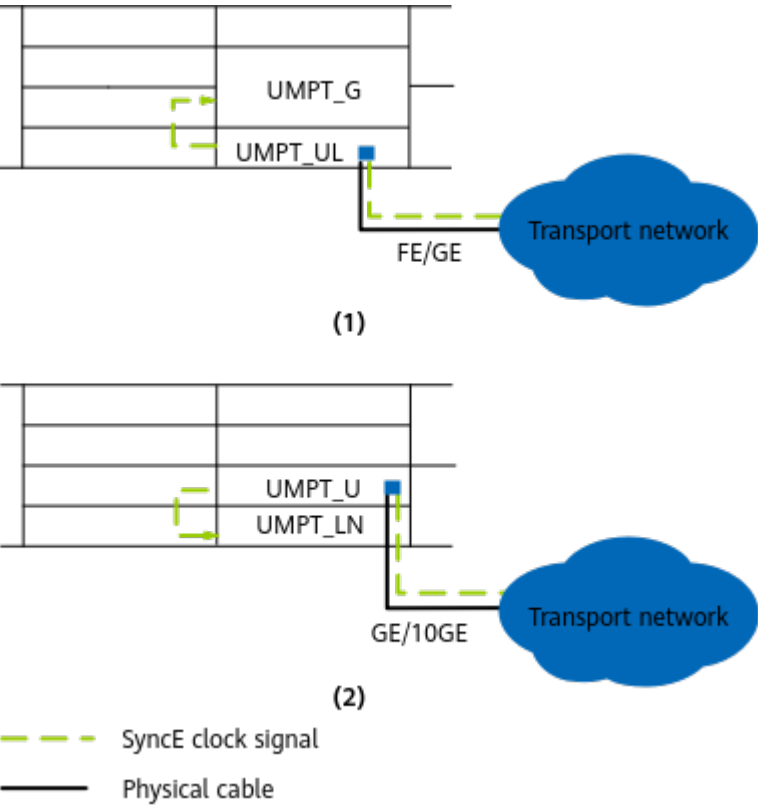


3.1.3.6.3 Common Synchronous Ethernet Clock in Hybrid-MPT Multimode Base Stations

Single BBU Scenarios

As shown in [Figure 3-32](#), only one BBU is used. RATs of a multimode base station share a synchronous Ethernet clock source through the BBU backplane.

Figure 3-32 Common synchronous Ethernet clock in a GUL hybrid-MPT base station



BBU Interconnection Scenarios

The common clock in BBU interconnection scenarios of hybrid-MPT base stations works in the same way as that in BBU interconnection scenarios of separate-MPT base stations. For details, see [BBU Interconnection Scenarios](#) in [3.1.3.6.1 Common Synchronous Ethernet Clock in Separate-MPT Multimode Base Stations](#).

3.1.3.7 Other Common Reference Clocks

In addition to common GPS/BeiDou/Galileo, BITS, E1/T1, IEEE1588 V2, and synchronous Ethernet reference clocks, the following common reference clocks can be used:

- Common 1PPS+TOD reference clock
- Common IEEE1588 V2+SyncE reference clock

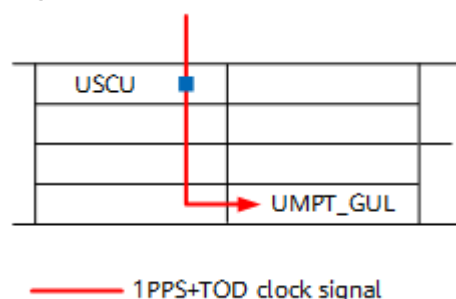
3.1.3.7.1 Common 1PPS+TOD Clock in Co-MPT Multimode Base Stations

The 1PPS+TOD clock source supports both frequency synchronization and time synchronization.

Single BBU Scenarios

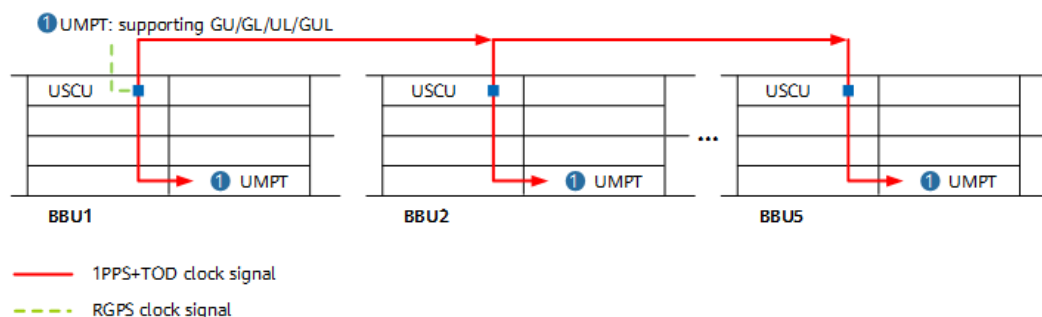
As shown in [Figure 3-33](#), the USCU receives 1PPS+TOD clock signals. The UMPT board is configured with the 1PPS+TOD clock source that is shared by GUL.

Figure 3-33 Common 1PPS+TOD clock in a GUL co-MPT multimode base station



The BBU is connected to the RGPS clock source through the USCUB, or to the GNSS clock source through the USCU. The USCU outputs 1PPS+TOD clock signals and transmits these clock signals to the main control board in the same BBU.

The 1PPS+TOD clock source provides either frequency synchronization or time synchronization for lower-level BBUs and, when the USCUB is used, also provides the clock signals for lower-level BBUs to implement inter-BBU clock sharing. A maximum of five cascaded BBUs are supported. The following figure shows an example.



The 1PPS+TOD clock source configurations for level-2 through level-5 BBUs are the same as those shown in [Figure 3-33](#). For details, see *Synchronization*.

BBU Interconnection Scenarios

The following uses the BBU interconnection achieved by connecting two UMPT boards as an example to describe the principle of clock sharing.

- As shown in [Figure 3-34](#), the 1PPS+TOD clock is located in the primary BBU and can be obtained from the USCUs. If the USCUs that introduces 1PPS+TOD clock signals is a USCUB and is configured for both RATs, the two RATs in the primary BBU cannot both use time synchronization. One RAT can use time synchronization and provides clock signals to the other RAT.
- As shown in [Figure 3-35](#), the 1PPS+TOD clock is located in the secondary BBU and can be obtained from the USCUs.

Figure 3-34 Common 1PPS+TOD clock in the primary BBU of a GUL base station

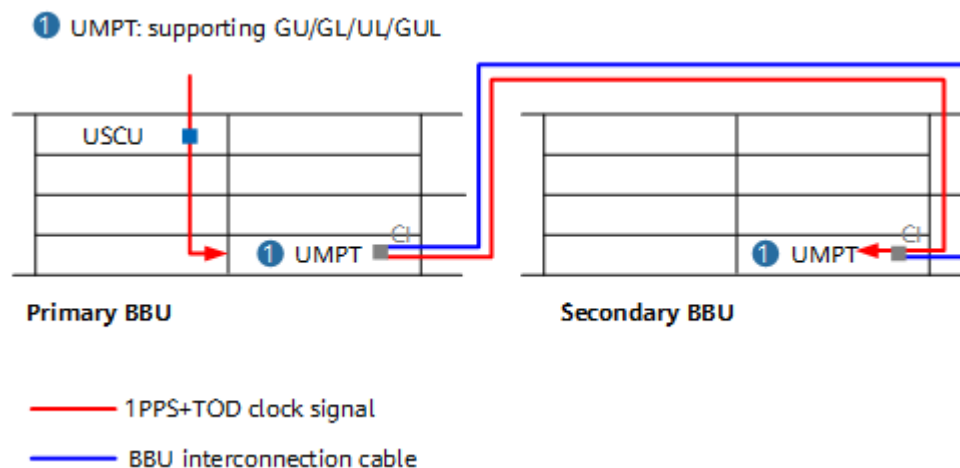
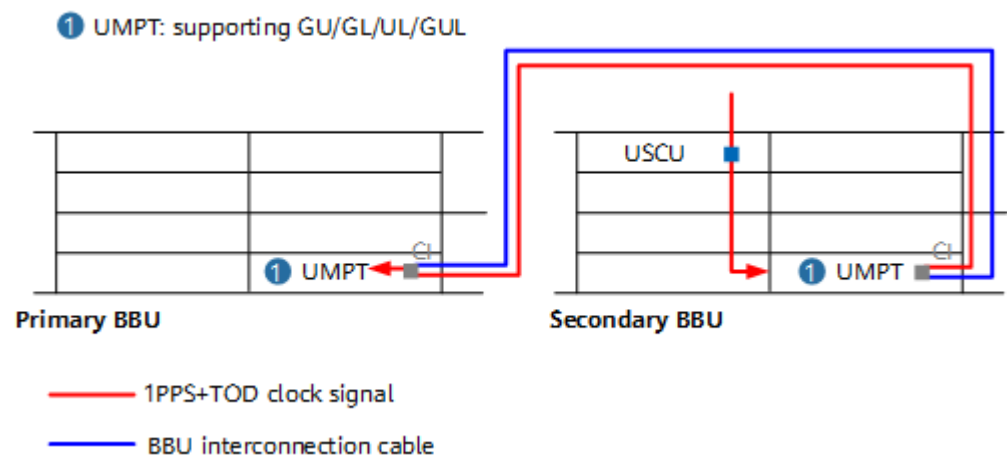


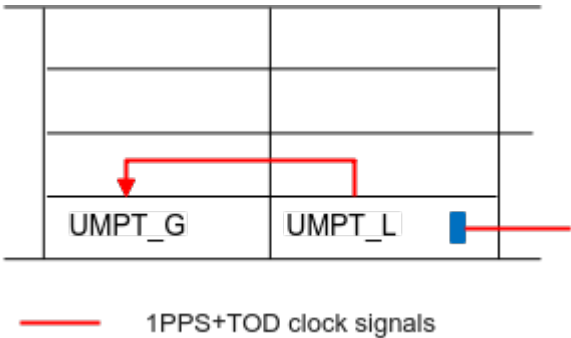
Figure 3-35 Common 1PPS+TOD clock in the secondary BBU of a GUL base station



3.1.3.7.2 Common 1PPS+TOD Clock in Separate-MPT Multimode Base Stations

In separate-MPT scenarios, when the UMPT receives 1PPS+TOD clock signals, the clock can be used as a shared clock source for other RATs.

Figure 3-36 Common 1PPS+TOD clock in a GL co-MPT multimode base station



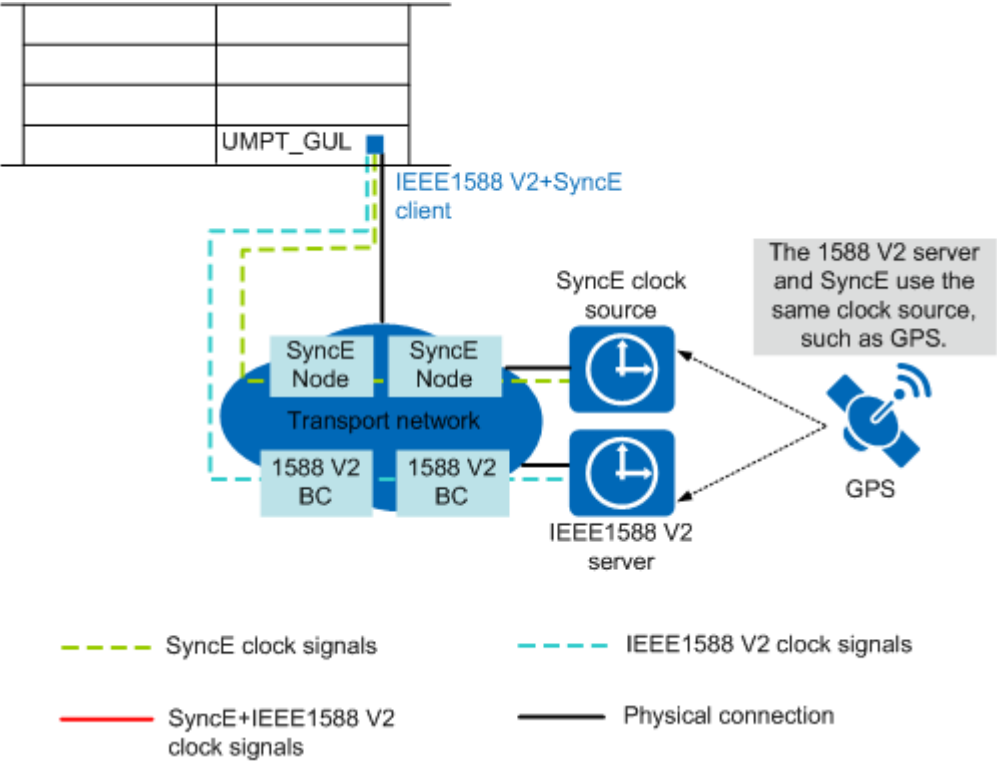
3.1.3.7.3 Common IEEE1588 V2+SyncE Clock in Co-MPT Multimode Base Stations

The IEEE1588 V2+SyncE clock source combines the IEEE1588 V2 clock with the synchronous Ethernet clock and is used for time synchronization. The clock source of the IEEE1588 V2 clock server must be the same as that of the synchronous Ethernet, such as GPS.

Single BBU Scenarios

As shown in [Figure 3-37](#), the UMPT is configured with the IEEE1588 V2+SyncE clock source that is shared by GSM, UMTS, and LTE.

Figure 3-37 Common IEEE1588 V2+SyncE clock in a GUL co-MPT base station



BBU Interconnection Scenarios

The following uses the BBU interconnection achieved by connecting two UMPT boards as an example to describe the principle of clock sharing.

- As shown in [Figure 3-38](#), the IEEE1588 V2+SyncE clock source is located in the primary BBU. The UMPT in the primary BBU is configured as an IEEE1588 V2+SyncE clock client to receive IEEE1588 V2+SyncE clock packets. Then, the UMPT in the primary BBU generates 1PPS clock signals that are synchronized with the server and forwards them to the secondary BBU through the BBU interconnection ports.
- As shown in [Figure 3-39](#), the IEEE1588 V2+SyncE clock source is located in the secondary BBU. The UMPT in the secondary BBU is configured as an IEEE1588 V2+SyncE clock client to receive IEEE1588 V2+SyncE clock packets. Then, the UMPT in the secondary BBU generates 1PPS clock signals that are synchronized with the server and forwards them to the primary BBU through the BBU interconnection ports.

Figure 3-38 Common IEEE1588 V2+SyncE clock in the primary BBU of a GUL base station

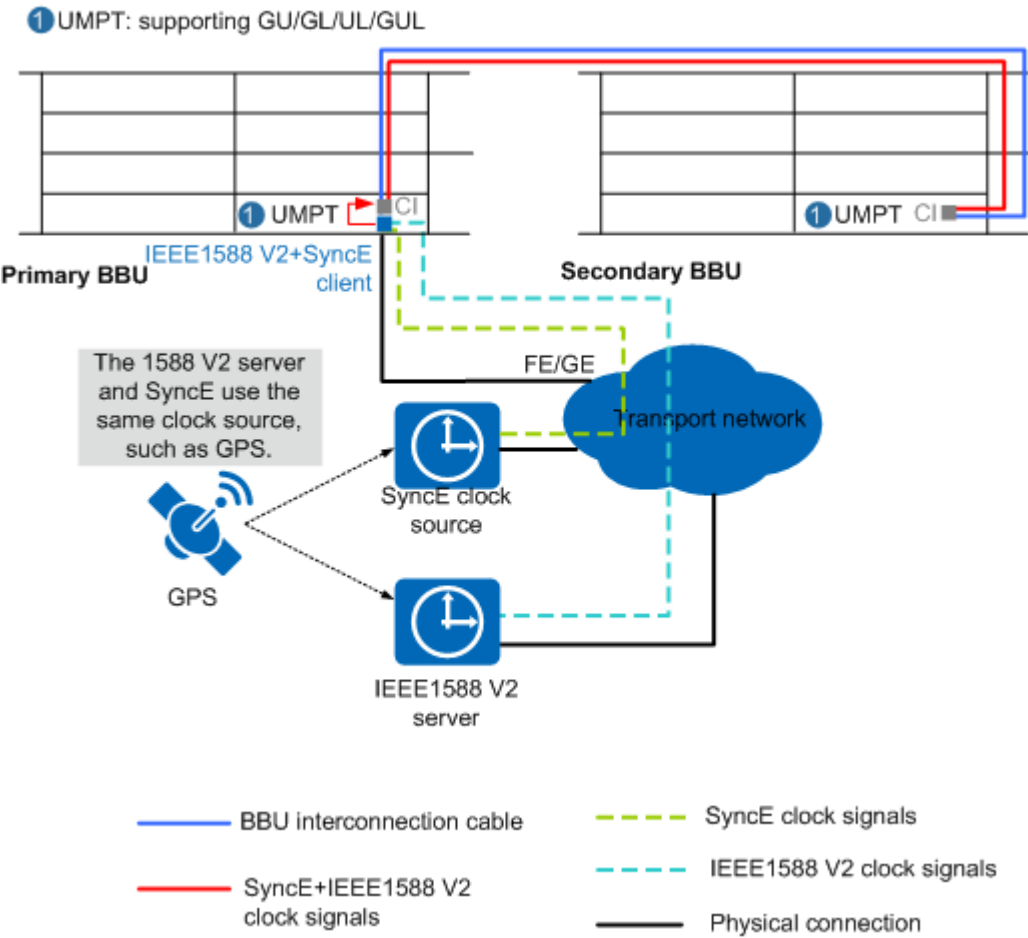
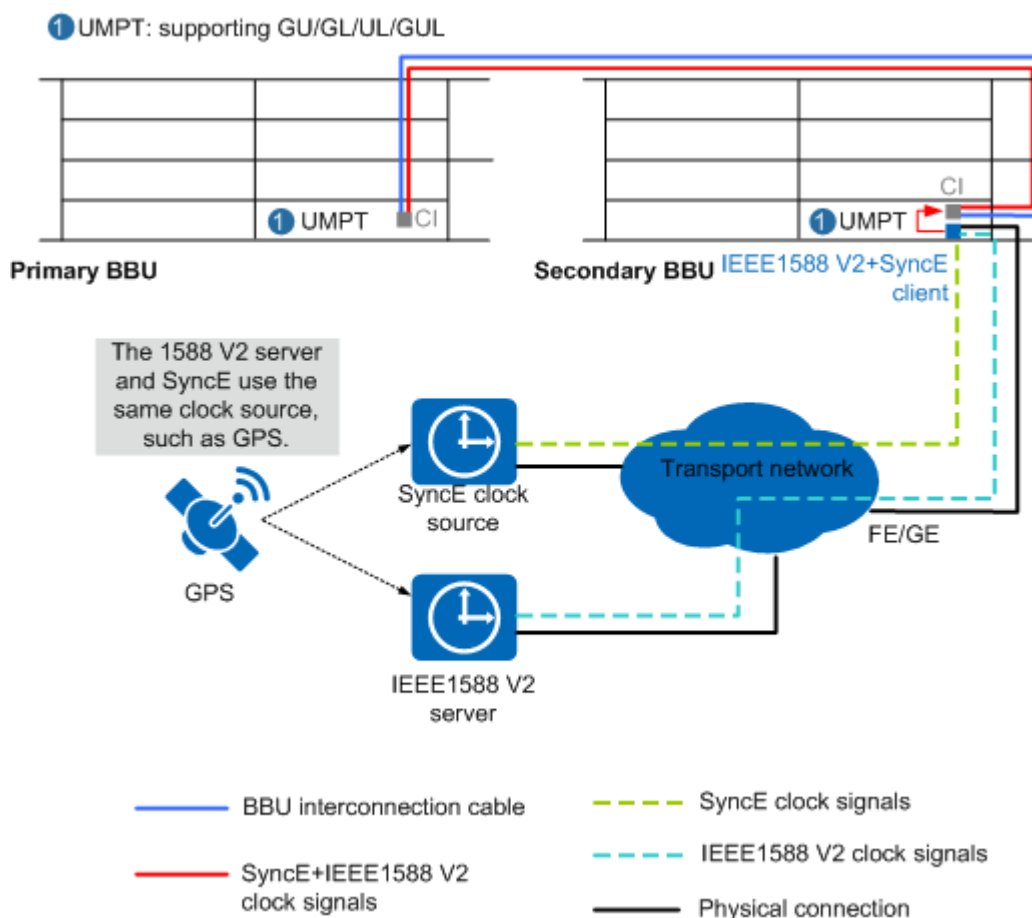
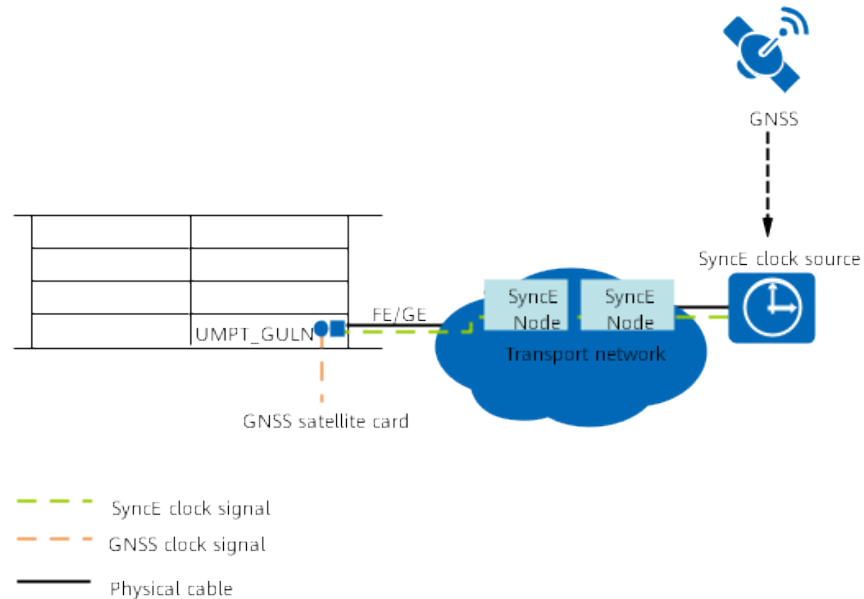


Figure 3-39 Common IEEE1588 V2+SyncE clock in the secondary BBU of a GUL base station



3.1.3.7.4 Common GNSS+SyncE Clock in Co-MPT Multimode Base Stations

The GNSS+SyncE clock source is a hybrid clock source and is used for time synchronization. Only LTE FDD supports this clock source combination. Therefore, the GNSS+SyncE reference clock can be shared only when GSM, UMTS, LTE FDD, and NR share one UMPT board. See [Figure 3-40](#).

Figure 3-40 Common GNSS+SyncE clock in a GULN co-MPT multimode base station

3.1.3.7.5 Common IEEE1588 V2+SyncE Clock in Separate-MPT Multimode Base Stations

The IEEE1588 V2+SyncE reference clock can provide time synchronization for G/U/L/N base stations. In separate-MPT scenarios, it can be used as the shared time synchronization source. In this case, the UMPT must be used to provide the shared IEEE1588 V2+SyncE clock source. There is no such restriction on the main control board that receives the shared clock signals.

In multimode scenarios, LTE and NR can be configured with different IEEE1588 V2+SyncE clock sources. This scenario has the following restrictions:

- If the LTE and NR modes are configured with different IEEE1588 V2+SyncE clock sources, they cannot use shared clock signals of the other mode.
- If one mode is configured with GNSS+IEEE1588 V2&SyncE clock backup, the other mode cannot be configured with IEEE1588 V2+SyncE clock source.

Single BBU Scenarios

As shown in [Figure 3-41](#), [Figure 3-42](#), [Figure 3-43](#), and [Figure 3-44](#), one mode is connected to the transport network through an FE/GE port. The RAT is configured as an IEEE1588 V2+SyncE clock client to receive clock signals. The local oscillator of the RAT synchronizes clock signals with the IEEE1588 V2+SyncE clock source and transmits the synchronized 1PPS clock signals to the other RAT through the backplane. In this case, only one IEEE1588 V2+SyncE clock client is required.

Figure 3-41 Common IEEE1588 V2+SyncE clock provided to GSM in a separate-MPT multimode base station

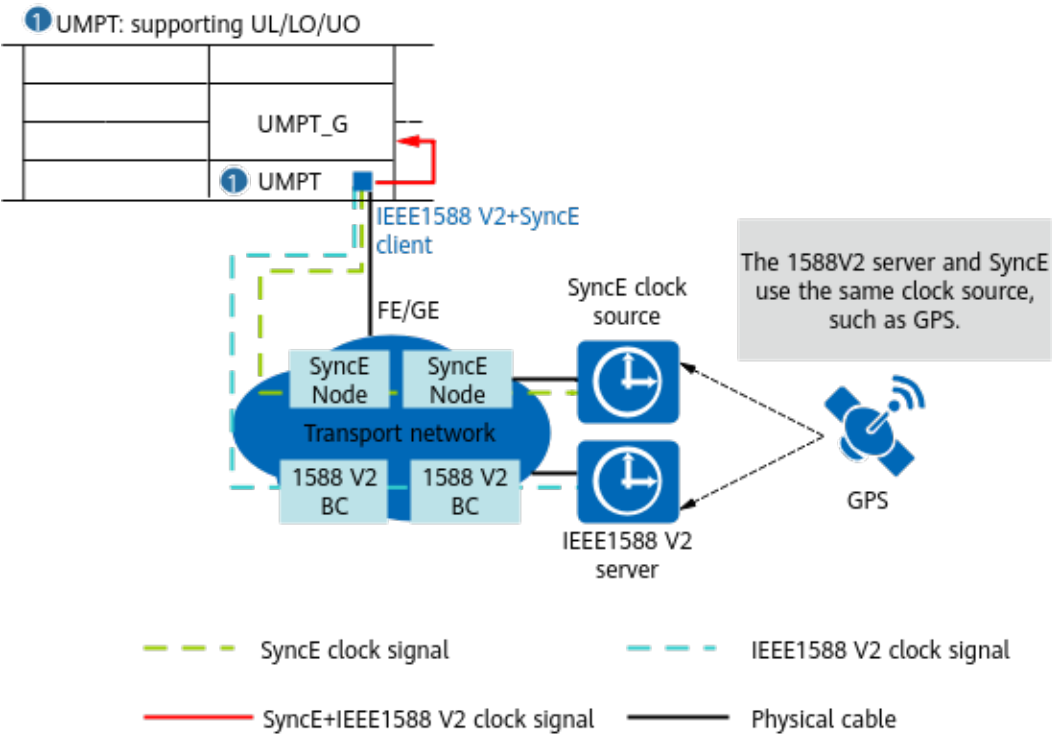


Figure 3-42 Common IEEE1588 V2+SyncE clock provided to UMTS in a separate-MPT multimode base station

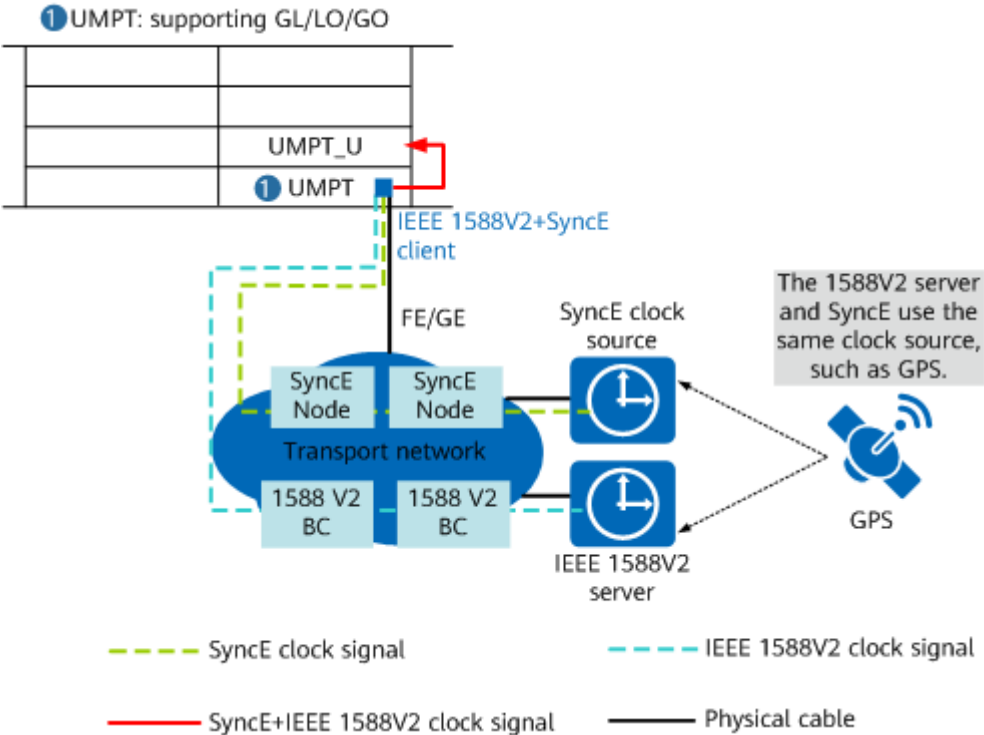


Figure 3-43 Common IEEE1588 V2+SyncE clock provided to LTE in a separate-MPT multimode base station

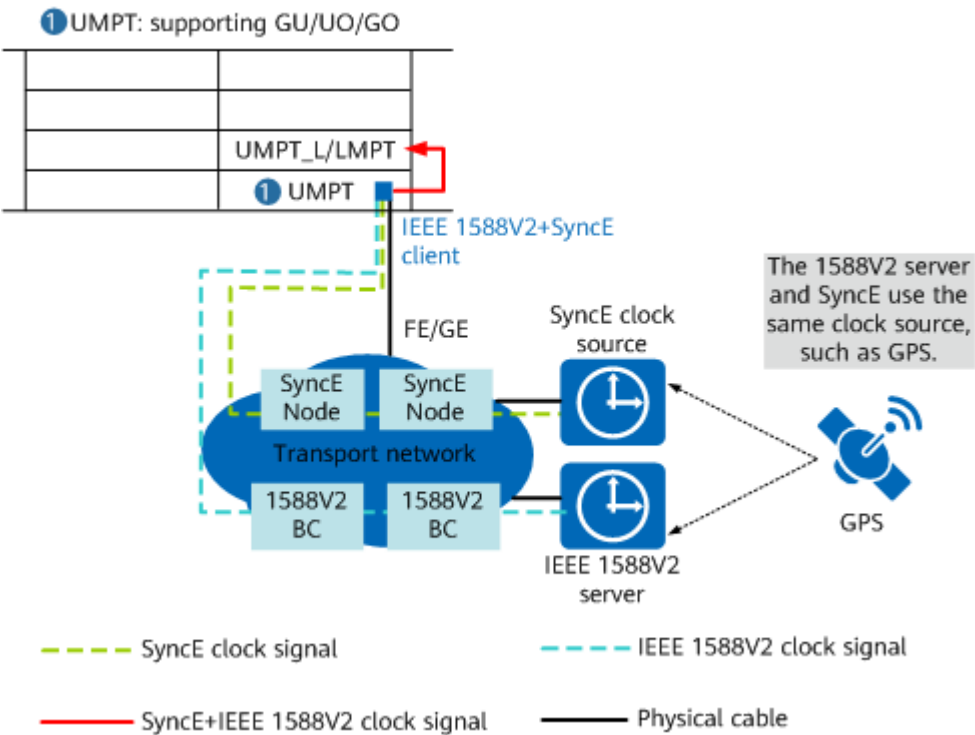
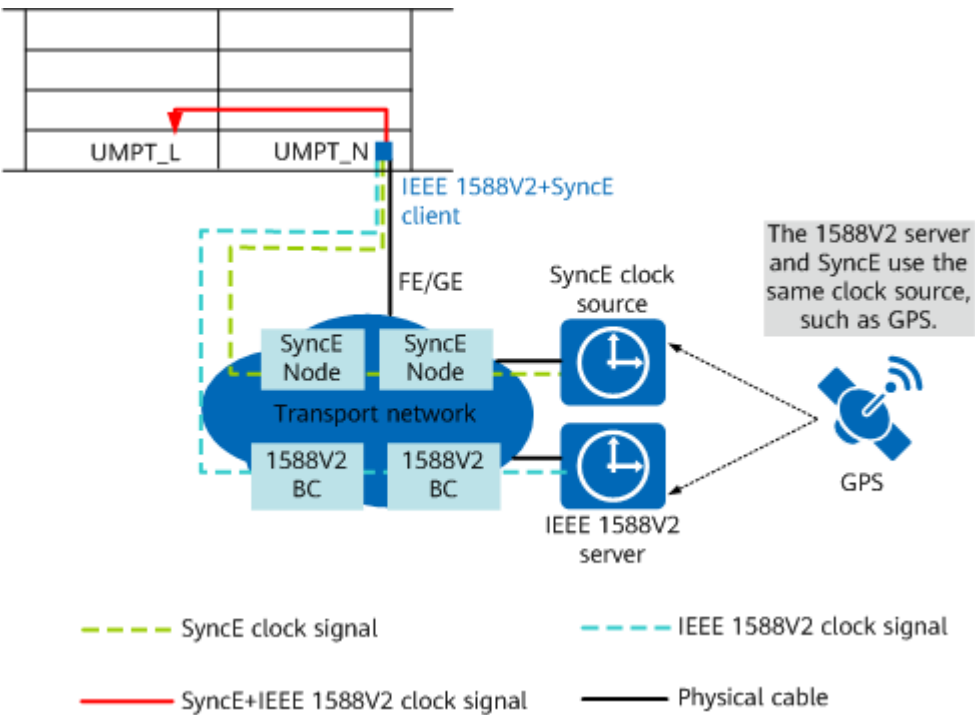


Figure 3-44 Common IEEE1588 V2+SyncE clock provided to NR in a separate-MPT multimode base station



BBU Interconnection Scenarios

The following uses the BBU interconnection achieved by connecting UMPTs as an example to describe the principle of clock sharing. One RAT is configured with an IEEE1588 V2+SyncE clock source and shares it with the other two RATs.

- As shown in [Figure 3-45](#), the IEEE1588 V2+SyncE clock source is located in the primary BBU. One RAT in the primary BBU is configured as an IEEE1588 V2+SyncE clock client to receive IEEE1588 V2+SyncE clock packets. Then, the RAT generates 1PPS clock signals that are synchronized with the server and forwards them to the secondary BBU through the BBU interconnection port of the UMPT. The other RAT in the primary BBU obtains 1PPS clock signals through the BBU backplane.
- As shown in [Figure 3-46](#), the IEEE1588 V2+SyncE clock source is located in the secondary BBU. The RAT in the secondary BBU is configured as an IEEE1588 V2+SyncE clock client to receive IEEE1588 V2+SyncE clock packets. Then, the RAT generates 1PPS clock signals that are synchronized with the server, and forwards them to the primary BBU through the BBU interconnection port on the UMPT. The RATs in the primary BBU use the 1PPS clock signals through the BBU backplane.

Figure 3-45 Common IEEE1588 V2+SyncE clock in the primary BBU of a GUL base station

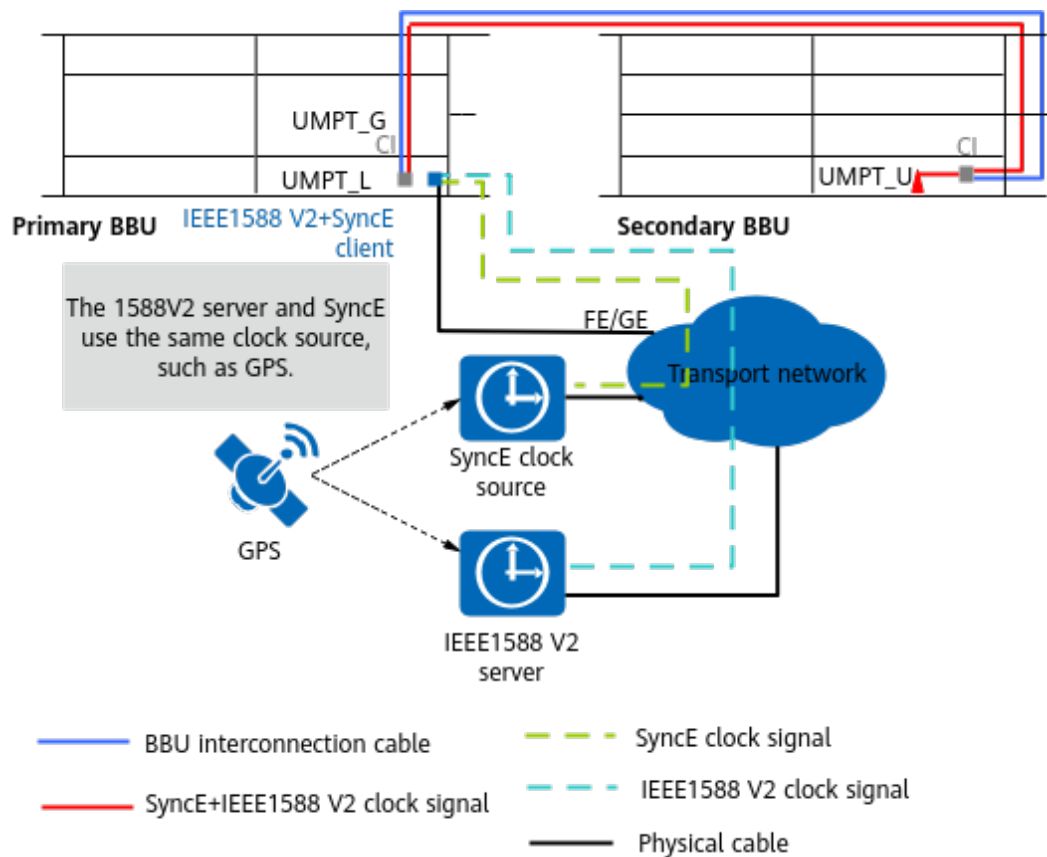
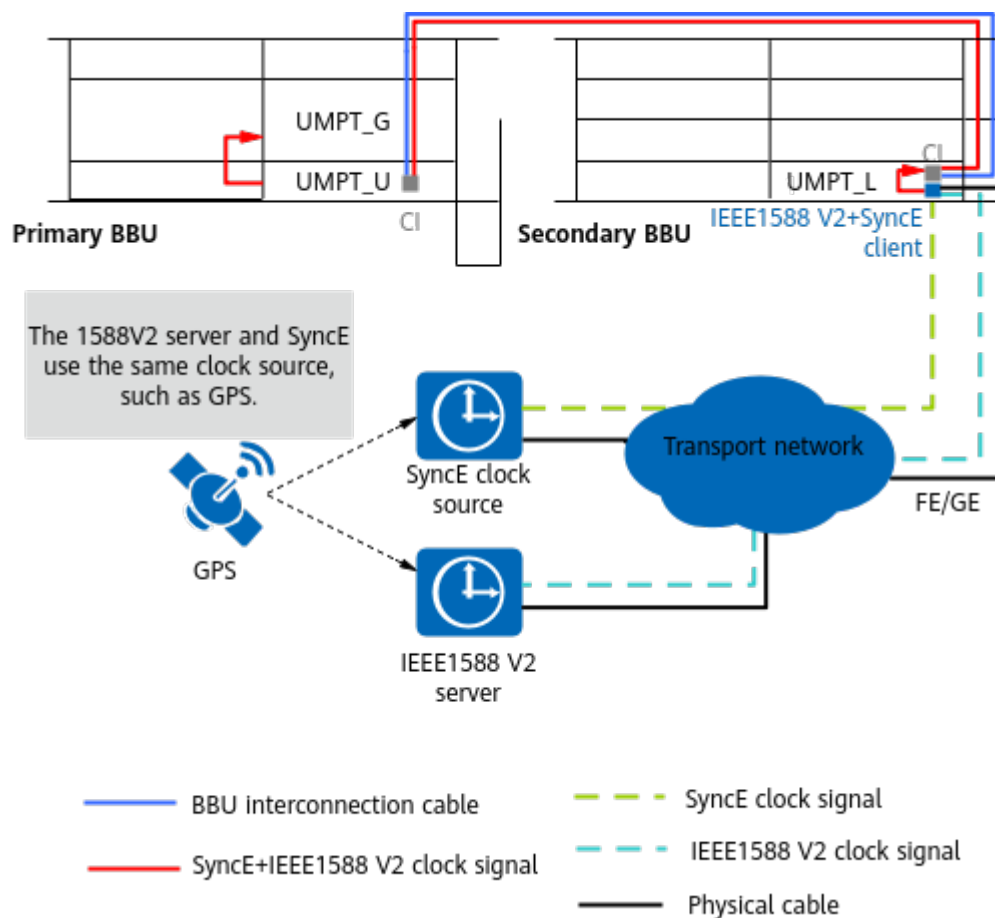


Figure 3-46 Common IEEE1588 V2+SyncE clock in the secondary BBU of a GUL base station



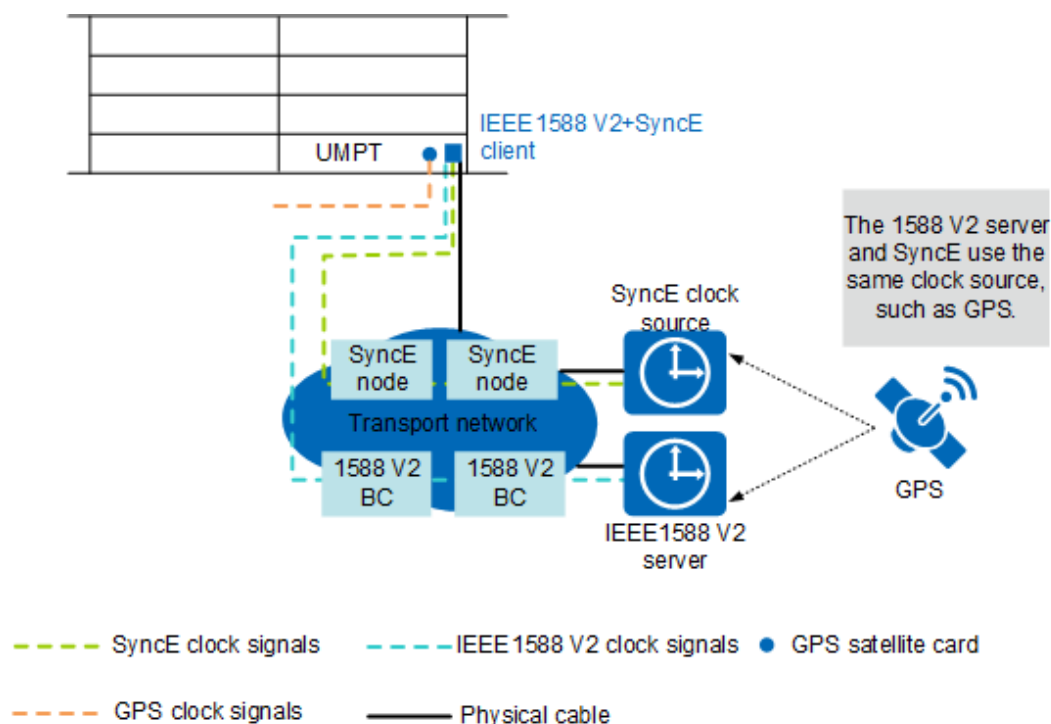
3.1.3.7.6 Common GNSS+IEEE1588 V2&SyncE Clock in Co-MPT Multimode Base Stations

The IEEE1588 V2&SyncE clock source is a hybrid clock source used for time synchronization. It works with GNSS to provide clock source backup. When the GNSS clock source is working, IEEE1588 V2&SyncE serves as the standby clock source and does not provide the clock source for the system clock. The multimode base station shares the GNSS+1588 V2&SyncE clock and must be configured with time synchronization.

Single BBU Scenarios

As shown in [Figure 3-47](#), the UMPT is configured with the GNSS+IEEE1588 V2&SyncE clock source, which is shared by all G/U/L/N mode.

Figure 3-47 Common GPS+IEEE1588 V2&SyncE clock in a GULN co-MPT base station



BBU Interconnection Scenarios

The following uses the BBU interconnection achieved by connecting two UMPT boards as an example to describe the principle of clock sharing.

- As shown in [Figure 3-48](#), when the GNSS+IEEE1588 V2&SyncE clock source is located in the primary BBU, the UMPT in the primary BBU synchronizes with the GNSS clock or terminates IEEE1588 V2&SyncE packets as a client. Then, the system generates 1PPS clock signals that are synchronized with the clock source and sends the 1PPS clock signals to the secondary BBU through the BBU interconnection port.
- As shown in [Figure 3-49](#), when the GNSS+IEEE1588 V2&SyncE clock source is located in the secondary BBU, the UMPT in the secondary BBU synchronizes with the GNSS clock or terminates IEEE1588 V2&SyncE packets as a client. Then, the system generates 1PPS clock signals that are synchronized with the clock source and sends the 1PPS clock signals to the primary BBU through the BBU interconnection port.

Figure 3-48 Common GNSS+IEEE1588 V2&SyncE clock in the primary BBU of a GULN base station

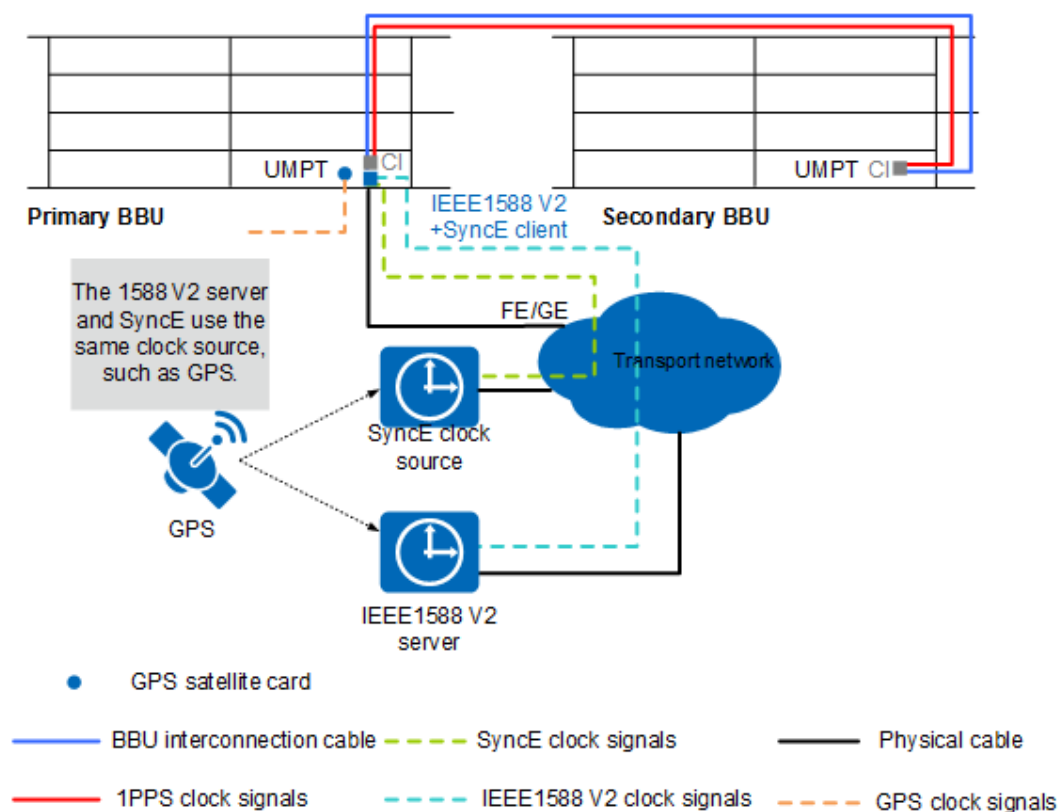
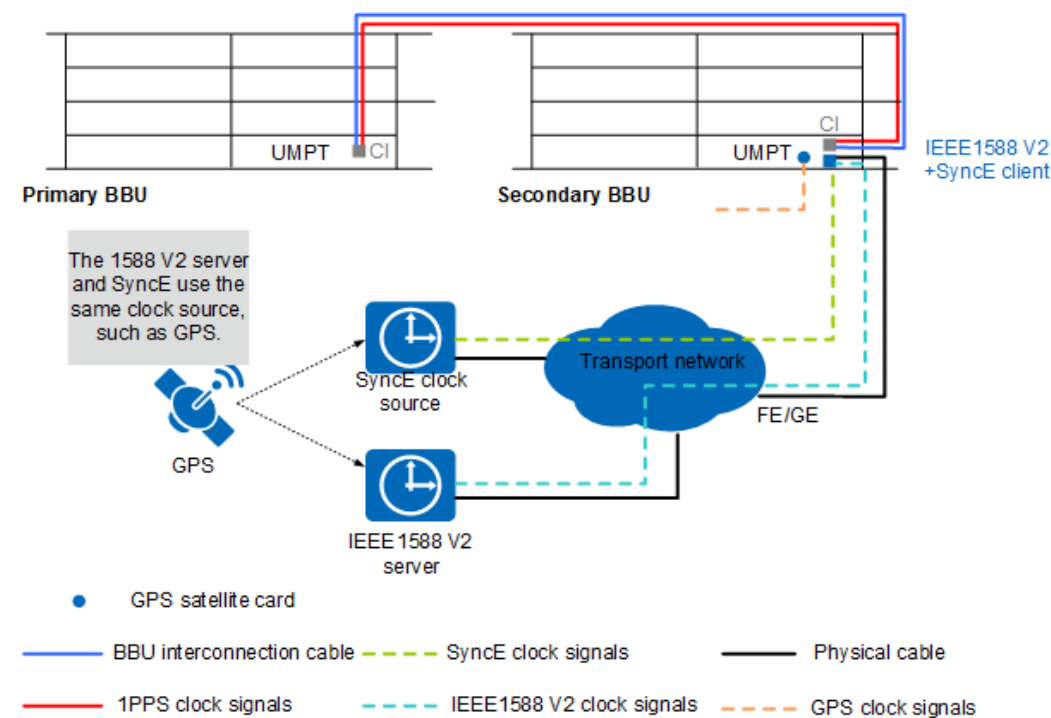


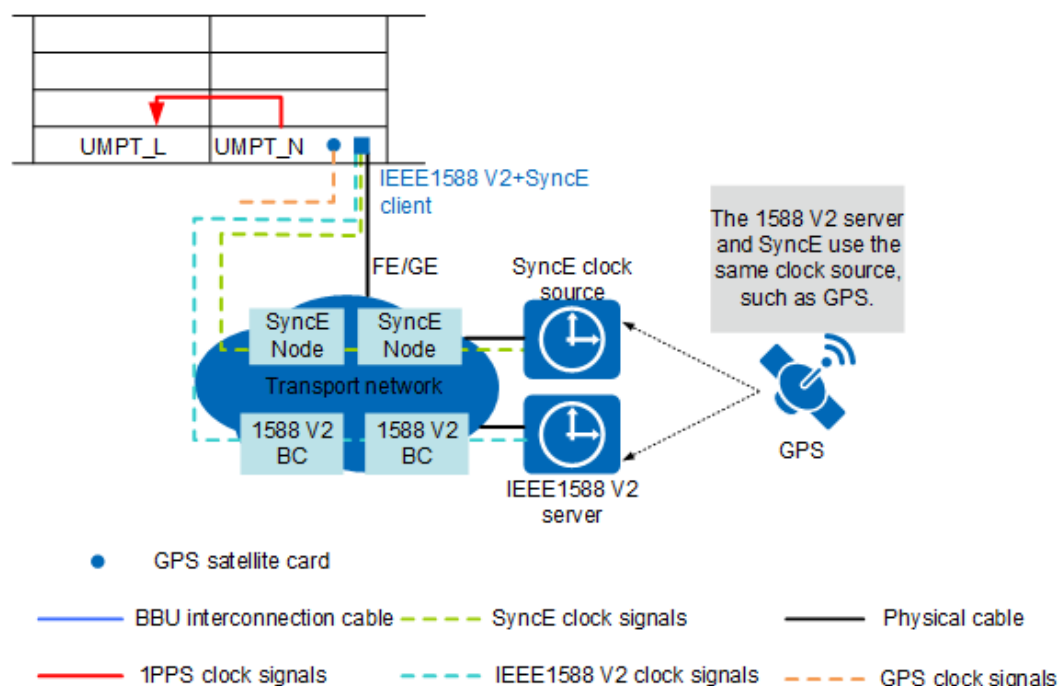
Figure 3-49 Common GNSS+IEEE1588 V2&SyncE clock in the secondary BBU of a GULN base station



The IEEE1588 V2&SyncE clock source is a hybrid clock source used for time synchronization. It works with GNSS to provide reference source backup. When the GNSS reference source is working, IEEE1588 V2&SyncE serves as the standby clock source and does not provide the reference source for the system clock. The multimode base station shares the GNSS+1588 V2&SyncE clock and must be configured with time synchronization. In separate-MPT scenarios, the UMPT must be used to provide the shared GNSS+IEEE1588 V2&SyncE clock source. There is no such restriction on the main control board that receives the shared clock signals. Only the LTE and NR main control boards support external GNSS+1588 V2&SyncE clock sources. There is no restriction on the receiver's modes.

In an LN separate-MPT base station, the gNodeB can synchronize with the GNSS clock or terminate IEEE1588 V2&SyncE packets as a client through the GNSS satellite card and FE/GE port, as shown in [Figure 3-50](#). The gNodeB generates 1PPS clock signals that are synchronized with the clock source and sends the 1PPS clock signals to the eNodeB through the backplane. The eNodeB uses the 1PPS clock signals as the reference clock source. In this scenario, only one mode needs to be configured with the GNSS+IEEE1588 V2&SyncE clock source.

Figure 3-50 Common GNSS+IEEE1588 V2&SyncE clock in an LN separate-MPT base station

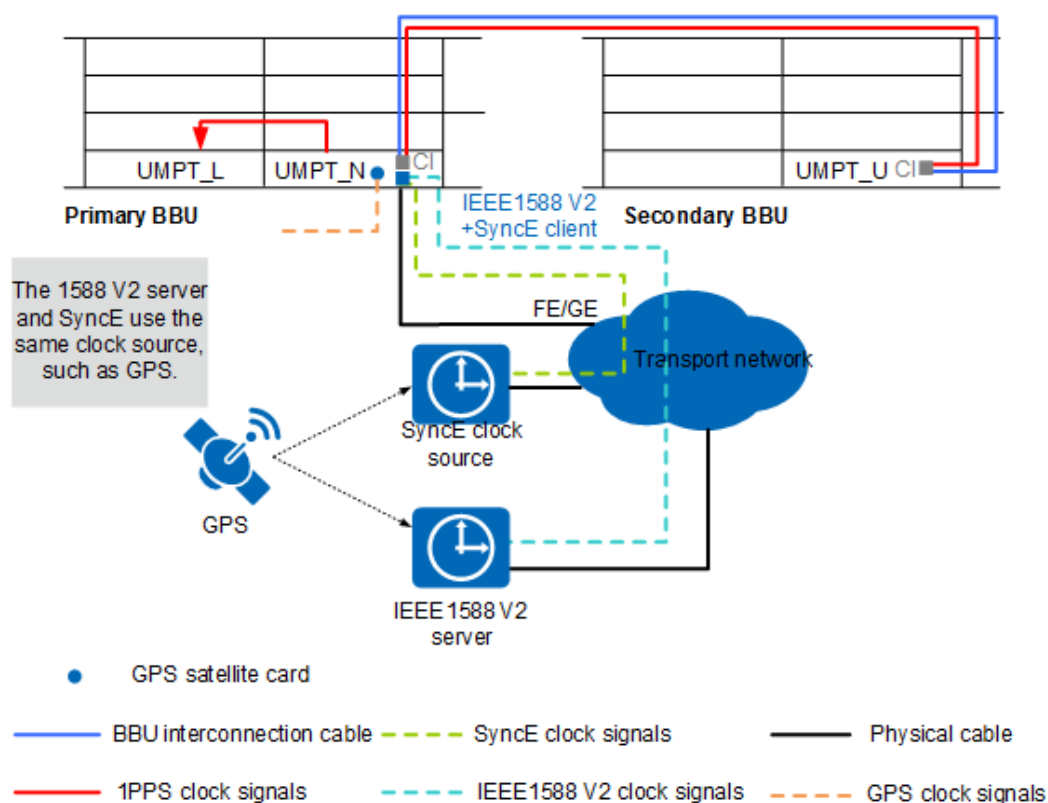


In this example, two UMPT boards are used to interconnect two BBUs. One mode is configured with the GNSS+IEEE1588 V2&SyncE clock source, and the other two modes use the shared GNSS+IEEE1588 V2&SyncE clock source. If one mode is

configured with the GNSS+IEEE1588 V2&SyncE clock source, other modes can only use the shared clock source but cannot be configured with this clock source. The clock sharing process is as follows:

- In an LN+U separate-MPT base station, as shown in [Figure 3-51](#), when the GNSS+IEEE1588 V2&SyncE clock source is located in the primary BBU, the RAT that accesses the GNSS+IEEE1588 V2&SyncE clock source is synchronized with the GNSS, or serves as an IEEE1588 V2&SyncE client to terminate IEEE1588 V2&SyncE packets, and generates synchronized 1PPS clock signals. Then, the 1PPS clock signals are sent to the secondary BBU through the BBU interconnection port on the UMPT. The other RAT in the primary BBU obtains 1PPS clock signals through the BBU backplane.
- In an LN+U separate-MPT base station, as shown in [Figure 3-52](#), when the GNSS+IEEE1588 V2&SyncE clock source is located in the secondary BBU, the RAT that accesses the GNSS+IEEE1588 V2&SyncE clock source is synchronized with the GNSS, or serves as an IEEE1588 V2&SyncE client to terminate IEEE1588 V2&SyncE packets, and generates synchronized 1PPS clock signals. Then, the 1PPS clock signals are sent to the primary BBU through the BBU interconnection port on the UMPT. The other RAT in the primary BBU obtains 1PPS clock signals through the BBU backplane.

Figure 3-51 Common GNSS+IEEE1588 V2&SyncE clock in the primary BBU of an LN+U base station



The diagram illustrates the clock distribution architecture between a Primary BBU and a Secondary BBU. At the top, two tables represent the clock distribution matrices for each BBU. The Primary BBU table has columns for UMPT_U, UMPT_L, and CI. The Secondary BBU table has columns for UMPT_N and CI. A red line indicates the 1PPS clock signal path from the CI of the Primary BBU to the CI of the Secondary BBU. A blue line indicates the SyncE clock signal path from the UMPT_N of the Secondary BBU to the CI of the Primary BBU. Below the tables, a text box states: "The 1588 V2 server and SyncE use the same clock source, such as GPS." A GPS satellite icon is shown connected to two clock sources: a "SyncE clock source" and an "IEEE 1588 V2 server". Both clock sources are connected to a "Transport network" cloud. The "SyncE clock source" is connected to the Primary BBU via a "SyncE clock signal" (dashed green line). The "IEEE 1588 V2 server" is connected to the Secondary BBU via an "IEEE 1588 V2 clock signal" (dashed cyan line). A "Physical cable" (solid black line) connects the Secondary BBU to the "FE/GE" port of the Primary BBU. A legend at the bottom defines the symbols: a blue dot for "GPS satellite card", a blue line for "BBU interconnection cable", a red line for "1PPS clock signals", a dashed green line for "SyncE clock signals", a dashed cyan line for "IEEE 1588 V2 clock signals", a solid black line for "Physical cable", and a dashed orange line for "GPS clock signals".

3.2.1 Benefits

3.2.2 Impacts

This function has no impact on the network.

3.3 Requirements

3.3.1 Licenses

None

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 Prerequisite Functions

MRFD-121117 Multi-mode BS Common Reference Clock(GBTS)

RAT	Function Name	Function Switch	Reference (GBSS Feature Documentation)	Description
GSM	BTS GPS Synchronization	None	<i>BTS GNSS Synchronization</i>	This feature is required when common GPS clock is used.
GSM	Synchronization with BeiDou	None	<i>Synchronization</i>	This feature is required when common BeiDou clock is used.
GSM	BTS/NodeB Clock	None	<i>Synchronization</i>	This feature is required when common BITS/IEEE1588 V2/1PPS+TOD clock is used.
GSM	Synchronous Ethernet	None	<i>Synchronization</i>	This feature is required when common synchronous Ethernet/IEEE1588 V2+SyncE clock is used.

MRFD-121127 Multi-mode BS Common Reference Clock(NodeB)

RAT	Function Name	Function Switch	Reference (RAN Feature Documentation)	Description
UMTS	BTS/NodeB Clock	None	<i>Synchronization</i>	This feature is required when common GPS/BeiDou/BITS/1PPS+TOD clock is used.
UMTS	Synchronous Ethernet	None	<i>Synchronization</i>	This feature is required when common synchronous Ethernet clock is used.
UMTS	Clock Sync on Ethernet in NodeB	None	<i>Synchronization</i>	This feature is required when common IEEE1588 V2/IEEE1588 V2+SyncE clock is used.

MRFD-121137 Multi-mode BS Common Reference Clock(eNodeB)

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Synchronization with GPS	None	<i>Synchronization</i>	This feature is required when common GPS clock is used.
LTE FDD	Synchronization with BeiDou	None	<i>Synchronization</i>	This feature is required when common BeiDou clock is used.
LTE FDD	Synchronization with BITS	None	<i>Synchronization</i>	This feature is required when common BITS clock is used.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Synchronization with Ethernet (ITU-T G.8261)	None	<i>Synchronization</i>	This feature is required when common synchronous Ethernet/ IEEE1588 V2+SyncE clock is used.
LTE FDD	IEEE1588 V2 Clock Synchronization	None	<i>Synchronization</i>	This feature is required when common IEEE1588 V2/ IEEE1588 V2+SyncE clock is used.
LTE FDD	Synchronization with 1PPS	None	<i>Synchronization</i>	This feature is required when common 1PPS +TOD clock is used.

MRFD-121147 Multi-mode BS Common Reference Clock(LTE TDD)

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Synchronization with GPS	None	<i>Synchronization</i>	This feature is required when common GPS clock is used.
LTE TDD	Synchronization with BeiDou	None	<i>Synchronization</i>	This feature is required when common BeiDou clock is used.
LTE TDD	IEEE1588 V2 Clock Synchronization	None	<i>Synchronization</i>	This feature is required when common IEEE1588 V2/ IEEE1588 V2+SyncE clock is used.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Synchronization with 1PPS	None	<i>Synchronization</i>	This feature is required when common 1PPS +TOD clock is used.

MRFD-121157 Multi-mode BS Common Reference Clock(NB-IoT)

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
NB-IoT	Synchronization	None	<i>Synchronization</i>	This feature is required when NB-IoT provides the shared clock to other RATs.

MRFD-151166 Multi-mode BS Common Reference Clock(NR)

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Synchronization	None	<i>Synchronization</i>	This feature is required when NR provides the shared clock to other RATs.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Network Synchronization	None	<i>Synchronization</i>	This feature is required when NR provides the shared clock to other RATs.

3.3.2.2 Mutually Exclusive Functions

None

3.3.2.3 Function Impacts

RA T	Func ti on Name	Function Switch	Reference	Description
NR FD D Lo w- freq uen cy NR TD D Hig h- freq uen cy NR TD D	eXn self-manag ement	BBP.EXFUNCSW	<i>eXn Self-Management</i> in 5G RAN feature document ation	When the COORD_OUTPUT_BACKHAUL_PT option of the BBP.EXFUNCSW parameter is selected, HEI ports on BBPs cannot be interconnected to share clock signals.

3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations (macro base stations)
UMTS	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

If common BITS clock is used, a USCU must be installed in the BBU.

If common GPS clock is used, the USCU, or UMPT/UMDU containing a GPS satellite card, must be installed in the BBU. For details, see the satellite card types supported by boards in *Hardware Description* of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

If common BeiDou clock is used, a board supporting and configured with the BeiDou satellite card must be installed in the BBU. For details, see the satellite card types supported by boards in *Hardware Description* of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

If common IEEE1588 V2+SyncE clock is used, the main control board configured with the shared clock source must be the UMPT.

When a USCUB is managed by multiple RATs and the clock synchronization mode needs to be time synchronization, only one RAT can use time synchronization at a time.

RF Modules

This function does not depend on RF modules.

3.3.4 Others

Requirements for LTE TDD are as follows:

- The TDD eNodeB can only use the GPS clock provided by the GBTS that uses time synchronization. Other types of clock sources are not supported.
- The TDD eNodeB can only use the GPS/BeiDou clock source provided by the eGBTS/NodeB/gNodeB/co-MPT base station that uses time synchronization. Other types of clock sources are not supported.
- If a multimode base station includes LTE TDD, a clock source that supports time synchronization must be used, for example, GPS clock, BeiDou clock, IEEE1588 V2 clock, or 1PPS+TOD clock. In addition, the clock source must use time synchronization.

Requirements for NR TDD are as follows:

- A gNodeB can share only the GPS or BeiDou clock source provided by an eNodeB that uses time synchronization. Other types of clock sources are not supported.
- If a multimode base station includes NR TDD, a clock source that supports time synchronization must be used, for example, GPS clock, BeiDou clock, IEEE1588 V2 clock, or 1PPS+TOD clock. In addition, the clock source must use time synchronization.

3.4 Operation and Maintenance

3.4.1 When to Use

It is recommended that the Common Clock feature be activated for co-site multimode base stations to reduce the OPEX and CAPEX.

3.4.2 Precautions

For the GBTS/eGBTS, pay attention to the following points if the GPS, BeiDou, or IEEE1588 V2 clock source is used, regardless of whether the GBTS/eGBTS shares the clock source with other modes or uses the clock source provided by other modes:

- When the GBTS uses time synchronization, if the GBTS does not require inter-site frame number synchronization, set **BTSClk.FRAMESYNCSW** to **OFF**. If the GBTS requires inter-site frame number synchronization, set **BTSClk.FRAMESYNCSW** to **ON**. Before setting **BTSClk.FRAMESYNCSW** to **ON**, plan FN offset and TSC for GSM to avoid interference between neighboring GSM cells. The recommended value is **OFF**.
- When the eGBTS uses time synchronization, if the eGBTS does not require inter-site frame number synchronization, run the **SET CLKSYNCMODE** command with **TASM.GSMFRAMESYNCSW** set to **OFF**. If the eGBTS requires inter-site frame number synchronization, run the **SET CLKSYNCMODE** command with **TASM.GSMFRAMESYNCSW** set to **ON**. Before setting **TASM.GSMFRAMESYNCSW** to **ON**, plan FN offset and TSC for GSM to avoid interference between neighboring GSM cells. The recommended value is **OFF**.
- When a co-MPT base station involves GSM and GSM uses time synchronization, if GSM does not require inter-site frame number synchronization, run the **SET CLKSYNCMODE** command with **TASM.GSMFRAMESYNCSW** set to **OFF**. If GSM requires inter-site frame

number synchronization, run the **SET CLKSYNCMODE** command with **TASM.GSMFRAMESYNCSW** set to **ON**. Before setting **TASM.GSMFRAMESYNCSW** to **ON**, plan FN offset and TSC for GSM to avoid interference between neighboring GSM cells. The recommended value is **OFF**.

For details about GBTS/eGBTS inter-site frame number synchronization, see *Synchronization in GBSS Feature Documentation*.

3.4.3 Data Configuration

This document provides system clock configuration for each of the NEs that share a clock source and their common clock configuration. For details about how to configure the clock source of the NE that provides the clock source, see *Synchronization* for GSM BSS, WCDMA RAN, eRAN, or 5G RAN. This section describes how to configure a common clock in typical application scenarios.

3.4.3.1 Side Providing the Clock Source

A separate-MPT base station can use the following shared clock sources: GPS, BeiDou, BITS, E1/T1, IEEE1588 V2, synchronous Ethernet, and IEEE1588 V2+SyncE.

When an eGBTS/NodeB/eNodeB/gNodeB/co-MPT base station provides the clock source, they have the same data configuration of the clock source. The GBTS does not support the BeiDou or IEEE1588 V2+SyncE reference clock because its main control board cannot be the UMPT. Additionally, the configuration of the clock source (any of the clock sources listed above except BeiDou and IEEE1588 V2+SyncE) for a GBTS must be performed on the BSC. Therefore, the clock source data configuration for a GBTS is different from that for an eGBTS/NodeB/eNodeB/gNodeB/co-MPT base station. For details, see [3.4.3.1.2 GBTS Providing the Clock Source](#).

3.4.3.1.1 eGBTS/NodeB/eNodeB/gNodeB/Co-MPT Base Station Providing the Clock Source

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Configuring a GPS/BeiDou/Galileo Reference Clock

Step 1 Run the **ADD GNSS** command to add a GNSS clock link.

Parameter Name	Parameter ID	Setting Notes
GNSS Clock No.	GNSS.GN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
GNSS Work Mode	GNSS.MODE (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the type of the used satellite card. For example, if the satellite card is a single-mode GPS satellite card, set this parameter to GPS .

Step 2 Run the **SET CLKMODE** command to activate the GNSS reference clock. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	TASM.MODE (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to MANUAL(Manual) .
Selected Clock Source	TASM.CLKSR <i>C (LTE eNodeB, 5G gNodeB)</i>	Set this parameter to GNSS(GNSS Clock) .

----End

Configuring Common BITS Clock

Step 1 Run the **ADD BITS** command to add a BITS clock link.

Step 2 Run the **SET CLKMODE** command to activate the BITS reference clock. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	TASM.MODE (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to MANUAL(Manual) .
Selected Clock Source	TASM.CLKSR <i>C (LTE eNodeB, 5G gNodeB)</i>	Set this parameter to BITS(BITS Clock) .

----End

Configuring Common E1/T1 Clock

Step 1 Run the **ADD LINECLK** command to add an E1/T1 clock link.

Step 2 Run the **SET CLKMODE** command to activate the E1/T1 reference clock. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	TASM.MODE (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to MANUAL(Manual) .
Selected Clock Source	TASM.CLKSR <i>C (LTE eNodeB, 5G gNodeB)</i>	Set this parameter to LINECLK(Line Clock) .

----End

Configuring Common 1588 V2 Clock

Step 1 Run the **ADD IPCLKLINK** command to add a 1588 V2 clock link. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Protocol Type	IPCLKLNK.IC <i>PT (LTE eNodeB, 5G gNodeB)</i>	Set this parameter to PTP(PTP) .
Profile Type	IPCLKLNK.PR <i>OFIETYPE (LTE eNodeB, 5G gNodeB)</i>	Set this parameter according to the type of the 1588 V2 clock used on the live network.
Slot No.	IPCLKLNK.SN (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to the number of the slot that accommodates the IP clock link.
Clock Net Mode	IPCLKLNK.CN <i>M (LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.
Client IPv4	IPCLKLNK.CI <i>P (LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the IPv4 address of the client on the IP clock link.
Server IPv4	IPCLKLNK.SIP (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to the IPv4 address of the server on the IP clock link.
Client IPv6	IPCLKLNK.CI <i>PV6 (LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the IPv6 address of the client on the IP clock link.

Parameter Name	Parameter ID	Setting Notes
Client IPv6	IPCLKLNK.SIP V6 (LTE eNodeB, 5G gNodeB)	Set this parameter to the IPv6 address of the server on the IP clock link.

NOTE

If the 1588 V2 clock server is not on the same network segment as the eGBTS/NodeB/eNodeB/gNodeB/co-MPT base station, run the **ADD IPRT** command to configure a route between the 1588 V2 clock server and the eGBTS/NodeB/eNodeB/gNodeB/co-MPT base station.

Step 2 Run the **SET CLKMODE** command to activate the 1588 V2 reference clock. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	TASM.MODE (LTE eNodeB, 5G gNodeB)	Set this parameter to MANUAL(Manual) .
Selected Clock Source	TASM.CLKSR C (LTE eNodeB, 5G gNodeB)	Set this parameter to IPCLK(IP Clock) .
IP Clock Time Synchronization Mode	TASM.SYNM ODE (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
Adaptive Timing Recovery Switch	TASM.ATRSW (LTE eNodeB, 5G gNodeB)	Set this parameter to ON(On) when the 1588V2 ATR clock source is used.

----End

Configuring Common Synchronous Ethernet Clock

Step 1 Run the **ADD SYNCETH** command to add a synchronous Ethernet clock link. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Slot No.	SYNCETH.SN (LTE eNodeB, 5G gNodeB)	Set this parameter to the number of the slot that accommodates the synchronous Ethernet clock link.

Parameter Name	Parameter ID	Setting Notes
Port No.	SYNCETH.PN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the number of the port that accommodates the synchronous Ethernet clock link.
SSM Selection	SYNCETH.SSM <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.

Step 2 Run the **SET CLKMODE** command to activate the synchronous Ethernet reference clock. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	TASM.MODE <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to MANUAL(Manual) .
Selected Clock Source	TASM.CLKSRC <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to SYNCETH(SyncEth Clock) .

----End

Configuring Common IEEE1588 V2+SyncE Clock

Step 1 Run the **ADD SYNCETH** command to add a synchronous Ethernet clock link. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Slot No.	SYNCETH.SN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the number of the slot that accommodates the synchronous Ethernet clock link.
Port No.	SYNCETH.PN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the number of the port that accommodates the synchronous Ethernet clock link.
SSM Selection	SYNCETH.SSM <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.
SyncEth Clock No.	SYNCETH.LN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to 0 for the input clock link and 1 for the output clock link.

Step 2 Run the **ADD IPCLKLNK** command to add an IEEE1588 V2 clock link. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Protocol Type	IPCLKLNK.ICPT (LTE eNodeB, 5G gNodeB)	Set this parameter to PTP(PTP) .
Profile Type	IPCLKLNK.PROFILETYPE (LTE eNodeB, 5G gNodeB)	In most cases, set this parameter to 1588V2(1588V2) . If the IP clock server is a third-party device, it is recommended that this parameter be set to G.8265.1(G.8265.1) when a frequency synchronization source is used and be set to G.8275.1(G.8275.1) when a time synchronization source is used.
Slot No.	IPCLKLNK.SN (LTE eNodeB, 5G gNodeB)	Set this parameter to the number of the slot that accommodates the IP clock link.
Clock Net Mode	IPCLKLNK.CNM (LTE eNodeB, 5G gNodeB)	Set this parameter to L2_MULTICAST(L2 Multicast) .
Client IPv4	IPCLKLNK.CIP (LTE eNodeB, 5G gNodeB)	Set this parameter to the IPv4 address of the client on the IP clock link.
Server IPv4	IPCLKLNK.SIP (LTE eNodeB, 5G gNodeB)	Set this parameter to the IPv4 address of the server on the IP clock link.
Client IPv6	IPCLKLNK.CIPV6 (LTE eNodeB, 5G gNodeB)	Set this parameter to the IPv6 address of the client on the IP clock link.
Client IPv6	IPCLKLNK.SIPV6 (LTE eNodeB, 5G gNodeB)	Set this parameter to the IPv6 address of the server on the IP clock link.

NOTE

If the IEEE1588 V2 clock server is not on the same network segment as the eGBTS/NodeB/eNodeB/gNodeB/co-MPT base station, run the **ADD IPRT** command to configure a route between the IEEE1588 V2 clock server and the eGBTS/NodeB/eNodeB/gNodeB/co-MPT base station.

Step 3 Run the **SET CLKMODE** command to activate the IEEE1588 V2+SyncE reference clock. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	TASM.MODE <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to MANUAL(Manual) .
Selected Clock Source	TASM.CLKSRC <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to SYNCETH+IPCLK(SyncEth Clock+IP Clock) .
IP Clock Time Synchronization Mode	TASM.SYNMODE <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.

----End

3.4.3.1.2 GBTS Providing the Clock Source

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Configuring Common GPS Clock

Step 1 Run the **SET BTSUSCUBP** command to add a GPS clock link.

Step 2 Run the **SET BTSCCLK** command with *ClkType* set to **TRCGPS_CLK(Trace GPS Clock)**.

----End

Configuring Common BITS Clock

Step 1 Run the **SET BTSCCLK** command with *ClkType* set to **EXTSYN_CLK(External Sync clock)**.

----End

Configuring Common E1/T1 Clock

Step 1 Run the **SET BTSCLK** command with *ClkType* set to **TRCBSC_CLK**(Trace BSC Clock).

----End

Configuring Common IEEE1588 V2 Clock

Step 1 Run the **SET BTSCLK** command with *ClkType* set to **IP_TIME**(IP Clock).

Step 2 Run the **SET BTSIPCLKPARA** command to configure parameters for an IEEE1588 V2 clock link. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Clock Protocol Type	BTSIPCLKPAR A.CLKPRTTYP E	Set this parameter to PTP (PTP).
Clock Synchronization Mode	BTSIPCLKPAR A.SYNMode	Set this parameter based on the network plan.
Clock Topology Mode	BTSIPCLKPAR A.CLKTOPOM ODE	Set this parameter based on the network plan.
Clock Reference Source Redundancy	BTSIPCLKPAR A.ISCLKREDU CY	Set this parameter based on the network plan.
Clock Server 0 IP Address	BTSIPCLKPAR A.MASTERIPA DDR	Set this parameter to the IP address of clock server 0 on the IP clock link.
Clock Server 1 IP Address	BTSIPCLKPAR A.SLAVEIPAD DR	Set this parameter to the IP address of clock server 1 on the IP clock link.

----End

Configuring Common Synchronous Ethernet Clock

Step 1 Run the **SET BTSCLK** command with *ClkType* set to **SYNETH_CLK**(SynEth Clock). Run the **ADD SYNCETH** command to add a synchronous Ethernet clock link. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Slot No.	BTSCLK.SN	Set this parameter to the number of the slot that accommodates the synchronous Ethernet clock link.

Parameter Name	Parameter ID	Setting Notes
Port No.	BTSCLK.PN	Set this parameter to the number of the port that accommodates the synchronous Ethernet clock link.

----End

3.4.3.2 Sharing Clock Source Provided by Another RAT

3.4.3.2.1 eGBTS/NodeB/eNodeB/gNodeB/Co-MPT Base Station Sharing Clock Source Provided by Another RAT

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

When an eGBTS/NodeB/eNodeB/gNodeB/co-MPT base station uses the clock source provided by another RAT, their clock source data configurations are irrelevant to the clock source type. The configuration procedure is as follows:

Step 1 Run the **ADD PEERCLK** command to add a peer reference clock link.

Parameter Name	Parameter ID	Setting Notes
Peer Clock Node Type	PEERCLK.PN <i>T (LTE eNodeB, 5G gNodeB)</i>	Set this parameter to BBU(BBU) .
Peer Standard	PEERCLK.PS <i>(LTE eNodeB, 5G gNodeB)</i>	Set the Peer Standard as follows: - The value DEFAULT indicates the other standard in the same BBU. - The value COMM or COMM2 indicates that the peer standard is co-MPT.

Step 2 Run the **SET CLKMODE** command to configure the working mode of the reference clock.

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	TASM.MODE (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to MANUAL(Manual) .
Selected Clock Source	TASM.CLKSR C (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to PEERCLK(Peer Clock) .

Step 3 Run the **SET CLKSYNCMODE** command to configure the clock synchronization mode of the RAT that uses the clock source provided by another RAT.

Parameter Name	Parameter ID	Setting Notes
Clock Synchronization Mode	TASM.CLKSY NCMODE (<i>LTE eNodeB, 5G gNodeB</i>)	When Clock Synchronization Mode is set to FREQ for the RAT that provides the clock source, this parameter must be set to FREQ for the RAT that uses the provided clock source. When Clock Synchronization Mode is set to TIME or HYBRID for the RAT that provides the clock source, the following are true: - If the RAT that provides the clock source shares an RF module with the RAT that uses the provided clock source and the RAT that uses the provided clock source is the major RAT in forcible clock lock, Clock Synchronization Mode must be set to TIME for the RAT that uses the provided clock source. - In other cases, this parameter can be set to either TIME or FREQ as required.

----End

3.4.3.2.2 GBTS Sharing Clock Source Provided by Another RAT

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

When a GBTS shares the clock source provided by another RAT, its clock source data configuration is irrelevant to the clock source type. The configuration procedure is as follows:

Step 1 Run the **SET BTSClk** command.

Set **ClkType** to **PEER_CLK(Peer Clock)**.

Set **STANDARD** to the value of **Peer Standard** queried through the **DSP BTSSTATUS** command.

Set **CLKSYNCMODE** according to the following requirements:

- When **Clock Synchronization Mode** is set to **FREQ** for the RAT that provides the clock source, this parameter must be set to **FREQ**.
- When **Clock Synchronization Mode** is set to **TIME** or **HYBRID** for the RAT that provides the clock source, the following are true:
 - If the GBTS shares an RF module with the RAT that provides the clock source or the GBTS and the RAT are deployed in different BBUs, this parameter must be set to **TIME**.
 - In other cases, this parameter can be set to either **TIME** or **FREQ** as required for the GBTS.

----End

3.4.4 Activation Verification

Step 1 Check whether the Common Clock feature has been activated by running the **LST BTSClk** command for a GBTS and the **DSP CLKSTAT** command for a NodeB/eNodeB/eGBTS/gNodeB/co-MPT base station.

If...	Then...
The reference clock type is the same as the configured one	Configuring the reference clock succeeds.
The reference clock type is different from the configured one	Configuring the reference clock fails.

Step 2 Check whether any alarm is generated.

If...	Then...
ALM-26276 Inter-System Site-Level Configuration Conflict is generated with the value of Configuration Item being Clock Reference Type Configuration Conflict	The clock reference types conflict between RATs and configuring the reference clock fails.

----End

3.4.5 Network Monitoring

If the following alarm is reported on the eGBTS/NodeB/eNodeB/gNodeB, the alarm causes are as follows.

Alarm ID	Alarm Name
26276	Inter-System Site-Level Configuration Conflict

- The reference clock types conflict. That is, the current RAT does not support this type of reference clock. In this case, check the configuration file of the RAT that provides the clock source. Then change the conflicting reference clock to a reference clock that can be shared by the current RAT.
- The clock synchronization mode of the RAT that provides the clock source does not match that of the current RAT. In this case, change the clock synchronization mode of the RAT that provides the clock source to ensure that the clock synchronization modes match between the local and peer ends.
- The clock source provided by the RAT is faulty if the clock synchronization mode is set to **HYBRID** for the RAT that provides the clock source. In this case, check and rectify the faulty clock source.

If the shared clock source is correctly configured but the base station cannot lock the clock source within 30 minutes after startup, check for the clock source fault. For details, see "Possible Issues" in *Synchronization* for GSM/UMTS/LTE/NR.

4 Inter-RAT Clock Lock

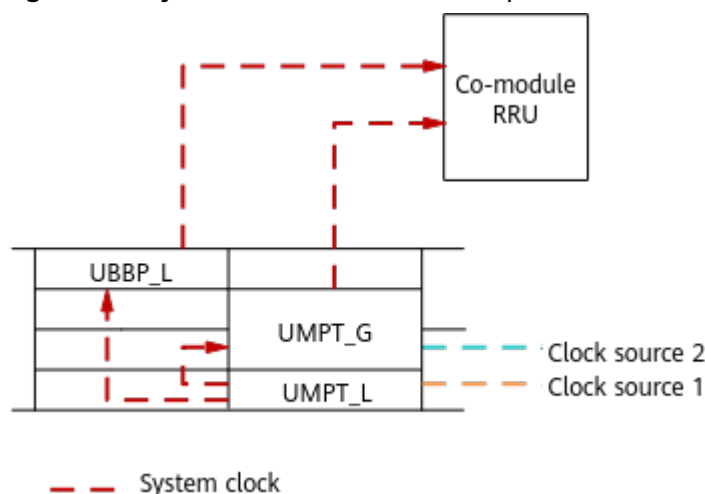
4.1 Principles

4.1.1 Overview

After a clock source is introduced to a base station, the clock system of the base station performs phase locking and frequency dividing on clock signals and then generates the system clock. The system clock is the working clock of the transmission, baseband, and RF subsystems of the base station.

If a separate-MPT multimode base station uses the dual-star or CPRI MUX networking and is enabled with the GU@5 MHz or GSM and LTE Spectrum Concurrency feature, the clock of each standard must strictly follow frequency synchronization and phase synchronization. That is, the system clocks among all RATs must be strictly synchronized. To achieve this, the system clocks among all RATs in a multimode base station must be mutually locked. After clock lock is enabled on the multimode base station, one RAT serves as the major standard. Upon receiving clock source signals, the major standard generates the system clock and provides the system clock for itself and other RATs. Other RATs are idle and do not provide clock reference for services of local RATs. Even if the minor-standard clock works in free-running mode, ongoing services are not affected. However, the reliability of the overall clock is affected. The major standard is negotiated automatically on a multimode base station and can be viewed using the **DSP CLKMUTUALLOCK** command. For example, on a GL separate-MPT base station, LTE serves as the major standard for clock lock and is connected to the RF module through the baseband processing unit. GSM serves as the minor standard, uses the system clock of LTE, and is directly connected to the RF module. See [Figure 4-1](#).

Figure 4-1 System clock lock in a GL separate-MPT base station



4.1.2 Clock Lock Conditions and Mode Changes

Clock Lock Conditions

If a separate-MPT multimode base station meets any of the following conditions, system clock lock is triggered:

- A multimode RF module is configured. (If the RAT synchronization mode is time synchronization, the external clock source must be locked.)
- The CPRI MUX feature is enabled. (If the RAT synchronization mode is time synchronization, the external clock source must be locked.)
- The GU@5 MHz feature is activated.
- The GSM and LTE Spectrum Concurrency and LTE and NR Spectrum Sharing features are activated.
- In a separate-MPT multimode base station, deployment is complete for GSM but only hardware installation is complete and deployment is not for the peer RAT.
- In BBU interconnection scenarios, if one RAT in the secondary subrack is in maintenance mode, the major standard for system clock lock is deployed in the primary subrack.

If a separate-MPT multimode base station meets none of the preceding conditions, system clock lock will not be triggered. Each RAT still uses its own system clock.

NOTE

For an RF module configured with multiple modes or enabled with the CPRI MUX feature, if clock lock is not enabled, the RRU/AAU randomly selects the system clock of one mode. If no clock of any mode is selected, UEs may fail to access the network. In addition, services on the multimode RF module are unavailable.

Clock Lock Mode Changes

On a separate-MPT multimode base station, a change of the clock lock mode is triggered in any of the following scenarios:

- The configurations that are related to clock lock are adjusted, such as the working standard of the RF module, and CPRI MUX. In this case, the separate-MPT multimode base station performs renegotiation according to the adjusted configurations. After the renegotiation, the minor standard switches from its own system clock (referred to as the local system clock) to the system clock of the major standard.
- The configurations that affect the decision of major and minor standards are adjusted. The configurations include system clock lock optimization switch and synchronization mode. (For selection of the major standard in clock lock, see [4.1.3.1 System Clock Lock of a Separate-MPT Base Station Including GBTS](#) and [4.1.3.2 System Clock Lock of a Separate-MPT Base Station Excluding GBTS](#).)
- All available clock sources of the major standard for clock lock are faulty and the minor standard has an available clock source. This may cause a switch from the system clock of the major standard to that of the minor standard. The duration from the time when the last clock source fault of the major standard occurs to the time when the switch is triggered varies depending on the type of the clock source that is the last to become faulty. The maximum duration is 8 hours.
- The major standard for clock lock is upgraded, reset, powered off, or removed and inserted. This causes the minor standard to switch back to the local system clock.
- Inter-subrack interconnection is interrupted. This causes the minor standard in the secondary BBU to switch back to the local system clock.
- The set of RATs that participate in clock lock is configured. This triggers clock lock between public and dedicated networks that are deployed in separate-MPT networking.

If the clock lock mode changes, services of the involved standards will be interrupted. When GSM serves as the major standard for clock lock, the interruption lasts no more than 8 minutes. When GSM does not serve as the major standard for clock lock, the interruption lasts no more than 3 minutes in intra-subrack scenarios and no more than 5 minutes in inter-subrack scenarios. When LTE or NR serves as the major standard for clock lock, the service interruption duration does not exceed 3 minutes. In intra-BBU scenarios, the service interruption duration does not exceed 1 minute. In inter-BBU scenarios, the service interruption duration does not exceed 3 minutes. During the clock lock mode switching, the following alarms may be generated for the involved standards. After the switching is complete, these alarms are cleared.

- ALM-26262 External Clock Reference Problem
- ALM-26264 System Clock Unlocked
- ALM-26265 Base Station Frame Number Synchronization Error
- ALM-26106 Board Clock Input Unavailable
- ALM-26234 BBU CPRI Interface Error
- ALM-26235 RF Unit Maintenance Link Failure
- ALM-26215 Inter-Board Service Link Failure
- ALM-26314 Inter-BBU Port Failure
- ALM-28302 Board BFN Abnormal

- ALM-26504 RF Unit CPRI Interface Error
- ALM-26538 RF Unit Clock Problem
- EVT-26361 Forcible Synchronization with Reference Clock Phase
- ALM-26324 BBU IR Interface Error
- ALM-26508 RF Unit IR Interface Error
- ALM-29870 NR DU Cell TRP Unavailable

 NOTE

In the preceding scenarios, the impacts of resource sharing such as common clock, co-transmission, and CPRI MUX are not considered and only the impacts of the system clock switching are considered.

4.1.3 Major Standard Selection in Clock Lock

In BBU interconnection scenarios, the major standard for clock lock must be located in the primary BBU.

When the RAT in the primary BBU works properly, the clock lock relationship is established. The major standard for clock lock is the RAT in the primary BBU. If the clock of the major standard for clock lock is not locked or the clock quality is poor and the minor standard for clock lock is LTE TDD or NR, the cell cannot be activated. To reduce the impact of the clock of the RAT in the primary BBU on the RAT in the secondary BBU, the following clock lock requirements must be met:

- If LTE TDD or NR is deployed in the secondary BBU and the clock in the primary BBU is normal and uses time synchronization, then the clock lock relationship is established between primary and secondary BBUs after the clock in the primary BBU is locked. If the clock in the primary BBU is abnormal for a long time, the clock lock relationship is established between primary and secondary BBUs 10 minutes after the attempt to establish the relationship is initiated.
- If LTE TDD or NR is deployed in the secondary BBU, the clock lock relationship is established between primary and secondary BBUs only when the RAT in the secondary BBU is not started (in maintenance mode and no external clock source is available). In other cases, the clock lock relationship is not established.

4.1.3.1 System Clock Lock of a Separate-MPT Base Station Including GBTS

If the system clock lock optimization function is not enabled for a separate-MPT base station including GBTS, GSM must serve as the major standard for clock lock because of the GTMU board capability restriction.

If the system clock lock optimization function is enabled and each RAT has been deployed, the major standard for clock lock is selected according to rules described in the following tables.

Single BBU Scenarios

Table 4-1 Major standard selection for clock lock of a GU/GL/G&U*L/G&L*N base station using a single BBU

Synchronization Mode (G)	Synchronization Mode (U/L)	Major Standard for Clock Lock
Frequency synchronization	Frequency synchronization	UMTS/LTE/UL/LN
Time synchronization	Time synchronization	UMTS/LTE/UL/LN
Frequency synchronization	Loose time synchronization	UMTS/LTE/UL/LN
Time synchronization	Frequency synchronization	GSM
Time synchronization	Loose time synchronization	GSM
Frequency synchronization ^a	Frequency synchronization	UMTS/LTE/UL/LN
Note: ^a indicates that the air interface soft synchronization function is enabled for the corresponding RAT. No mark indicates that the air interface soft synchronization function is not enabled for the corresponding RAT. If both soft-synchronized network and system clock lock optimization are enabled, system clock lock optimization takes effect as if soft-synchronized network is disabled.		

BBU Interconnection Scenarios

Table 4-2 Major standard selection for clock lock of a GU+L/GL+U/GU+LN (GSM in the primary BBU) base station using interconnected BBUs

Synchronization Mode (G)	Synchronization Mode (U/L in the Primary BBU)	Synchronization Mode (U/L/LN in the Secondary BBU)	Major Standard for Clock Lock
Frequency synchronization	Frequency synchronization	Frequency synchronization	UMTS/LTE in the primary BBU
Time synchronization	Time synchronization	Time synchronization	UMTS/LTE in the primary BBU
Frequency synchronization	Loose time synchronization	Loose time synchronization	UMTS/LTE in the primary BBU
Frequency synchronization	Loose time synchronization	Time synchronization	UMTS/LTE in the primary BBU

Synchronization Mode (G)	Synchronization Mode (U/L in the Primary BBU)	Synchronization Mode (U/L/LN in the Secondary BBU)	Major Standard for Clock Lock
Time synchronization	Frequency synchronization	Frequency synchronization	GSM in the primary BBU
Time synchronization	Loose time synchronization	Loose time synchronization	GSM in the primary BBU
Time synchronization	Loose time synchronization	Time synchronization	GSM in the primary BBU
Frequency synchronization ^a	Frequency synchronization	Frequency synchronization	UMTS/LTE in the primary BBU
Note: ^a indicates that the air interface soft synchronization function is enabled for the corresponding RAT. No mark indicates that the air interface soft synchronization function is not enabled for the corresponding RAT. If both soft-synchronized network and system clock lock optimization are enabled, system clock lock optimization takes effect as if soft-synchronized network is disabled.			

The following common clock deployment modes are recommended for a GL+N base station in which BBUs are interconnected, LTE serves as the major standard for clock lock, and the synchronization mode is time synchronization: (common GPS clock is used as an example)

- As shown in [Figure 4-2](#), the clock source is deployed on LTE in the primary BBU. GSM and NR use the clock signals provided by LTE. The secondary BBU uses the clock signals through the BBU interconnection ports on the UMPT boards. Co-SDR is used for GSM, LTE, and NR.
- As shown in [Figure 4-3](#), the clock source is deployed on NR in the secondary BBU. GSM and LTE use the clock signals provided by NR. The primary BBU uses the clock signals through the BBU interconnection ports on the UMPT boards. Co-SDR is used for GSM, LTE, and NR.

Figure 4-2 Common GPS clock in the primary BBU of a GL+N base station

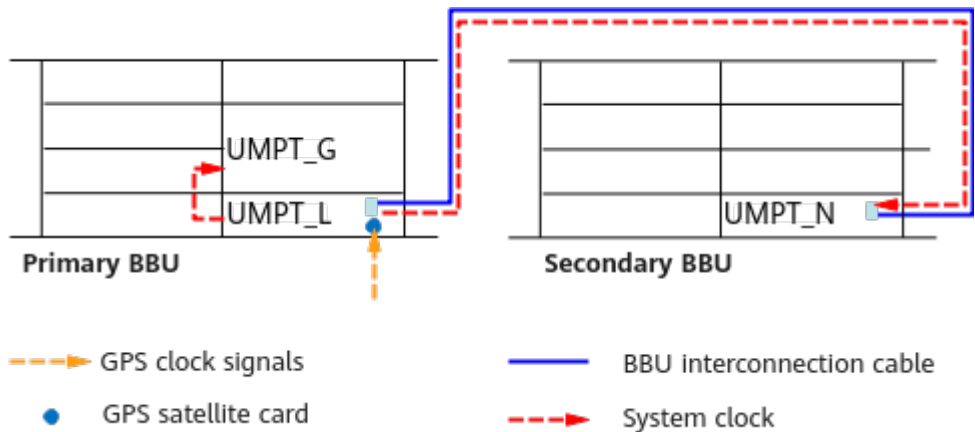


Figure 4-3 Common GPS clock in the secondary BBU of a GL+N base station

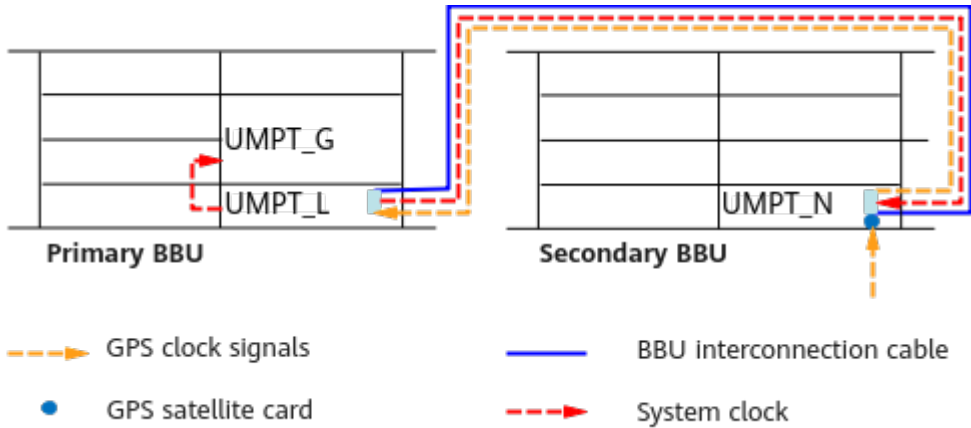


Table 4-3 Major standard selection for clock lock of a N+GL base station with BBUs interconnected (UMPT+UMPT interconnection, GL in the secondary BBU, and N in the primary BBU)

Synchronization Mode (N)	Synchronization Mode (L)	Synchronization Mode (G)	Major Standard for Clock Lock
Frequency synchronization	Frequency synchronization	Frequency synchronization	NR
Time synchronization	Time synchronization	Time synchronization	NR
Time synchronization	Frequency synchronization	Frequency synchronization	NR
Loose time synchronization	Frequency synchronization	Frequency synchronization	NR
Note: In this scenario, the air interface soft synchronization function is enabled for none of the RATs.			

NOTE

- The system clock lock optimization function is supported only when GSM uses the GTMUc and other RATs use the UMPT as the main control board.
- For details about the Uu based Soft Synchronization feature, see *Uu based Soft Synchronization* in *GBSS Feature Documentation*.
- NB-IoT uses low-precision frame synchronization to calculate the hyper frame number and frame number. In clock lock scenarios, if the RAT of NB-IoT is a minor standard, the calculated hyper frame number and frame number are based on the clock reference of the major standard. If frequency synchronization is configured for the major standard, the hyper frame number and frame number are incorrect. Therefore, it is recommended that the RAT of NB-IoT be configured as the major standard in the primary BBU. If a separate-MPT multimode base station includes GBTS, it is recommended that the system clock lock optimization feature be enabled for the GBTS.

4.1.3.2 System Clock Lock of a Separate-MPT Base Station Excluding GBTS

In a separate-MPT base station excluding GBTS with each RAT deployed: In single BBU scenarios, if each RAT uses the same synchronization mode and is not enabled with air interface soft synchronization or Cloud BB, the major standard for clock lock can be set using the **SET CLKMUTUALLOCK** command. The RAT whose system clock is unavailable cannot serve as the major standard for clock lock. For details, see [Table 4-4](#).

Table 4-4 Major standard selection for clock lock of a UL/NL base station using a single BBU

Synchronization Mode (U/N)	Synchronization Mode (L)	Whether Cloud BB Is Enabled for LTE	Major Standard for Clock Lock
Frequency synchronization	Frequency synchronization	No	Standard that is first activated
Time synchronization	Time synchronization	No	
Frequency synchronization	Frequency synchronization	Yes	LTE
Time synchronization	Time synchronization	Yes	
Frequency synchronization	Time synchronization	Yes/No	
Time synchronization	Frequency synchronization	Yes/No	UMTS/NR

NOTE

For details about Cloud BB, see *Cloud BB Overview* in *eRAN Feature Documentation*.

4.1.3.3 Forcible Major Standard for Clock Lock

If one main control board is faulty in the LTE and NR co-BBU separate-MPT networking, all services of the base station are affected. To solve this problem, the base station allows you to specify the major standard for clock lock that takes effect forcibly.

- In an LTE and NR co-BBU separate-MPT base station, if the synchronization modes of the two RATs are the same and neither of them is configured with the CGPS or RGPS clock source, the major standard for clock lock can be set to **LTE**, **NR**, or **AUTO** using the **TASM.FIXEDMS** parameter.
- For other types of base stations, this parameter must be set to **AUTO**. For details about how the major standard for clock lock is selected, see [4.1.3.1 System Clock Lock of a Separate-MPT Base Station Including GBTS](#) and [4.1.3.2 System Clock Lock of a Separate-MPT Base Station Excluding GBTS](#).

When the configuration of the forcible major standard for clock lock does not take effect, the base station reports ALM-26245 Configuration Data Inconsistency. If the forcible major standard for clock lock is configured on both the LTE and NR main control boards and the configured major standards are inconsistent, the earlier configuration takes effect on one base station, and ALM-26276 Inter-System Site-Level Configuration Conflict is reported on the other base station.

NOTE

When the **NODE.WMEXTENSION** parameter setting takes effect, the forcible major standard for clock lock cannot be set on the base station.

It is recommended that the function of configuring the forcible major standard for clock lock be enabled only when one of the two main control boards for clock lock is faulty. This prevents base station services from being affected by the faulty main control board. After the function of configuring the forcible major standard for clock lock is enabled, the base station no longer performs clock switching based on the clock status of the major and minor standards for clock lock. Therefore, clock switching is not triggered when the major standard for clock lock is unlocked.

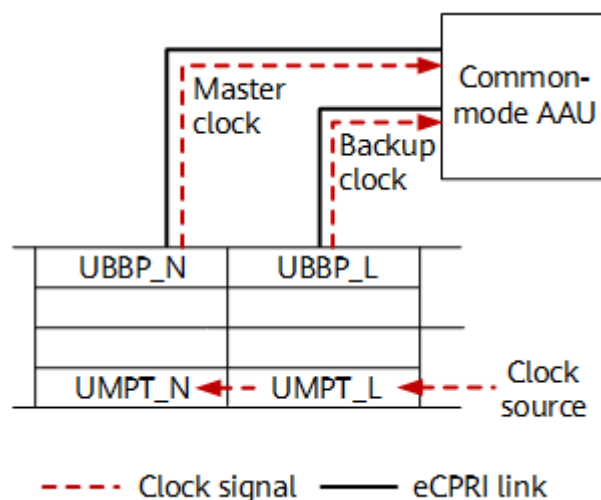
If both main control boards are normal, the value **AUTO** is recommended.

4.1.4 Clock Unlock in eCPRI Scenarios

Co-RF networking requires clock lock between modes of a multimode base station. However, the clock lock mechanism causes clock coupling between modes, causing the following problems:

- When clock lock is enabled for LTE and NR and LTE is the major standard, the precision and holdover capability of the NR clock decrease.
- The upgrade, reset, startup, or fault of any mode in a multimode base station affects other modes.

In eCPRI scenarios (currently, only LTE and NR support eCPRI interfaces), the clock single-source soft lock solution (clock unlock) can be used to resolve the preceding problems. After clock unlock is enabled, the clocks of different modes are synchronized with the reference clock. This ensures that the synchronization relationship between modes meets the synchronization requirement that the delay difference between services of different modes is within a certain range. In addition, the clocks of different modes are decoupled. Each mode obtains its own system clock. The multimode AAU selects the clock of one mode as the master clock and the clock of the other mode as the backup clock. When the master clock is faulty, the backup clock is used, as shown in [Figure 4-4](#). This function supports both co-BBU and inter-BBU scenarios. In inter-BBU interconnection scenarios, the clock does not depend on inter-BBU CI interconnection.

Figure 4-4 Clock unlock in eCPRI scenarios

The **TASM.ECPRIDEMULLOCKSW** parameter must be set to **ON** to enable this function. This function is supported only when all the following conditions are met:

- The clock source must be GNSS, 1588 V2, or GNSS+1588 V2. It is recommended that LTE or NR share the external clock source to reduce the synchronization deviation between RATs. The RGPS clock does not support this function.
- The **TASM.CLKSYNCMODE** parameter must be set to **TIME**.
- eCPRI RF modules can be shared only in LTE TDD and NR TDD, and the dual-star topology must be used.
- The clock unlock switch is turned on and the time synchronization reference source is connected for both LTE and NR that support eCPRI clock unlock.

eCPRI clock unlock solves the clock coupling problem between RATs. However, the clock quality of the peer end cannot be detected because the clocks of different RATs are independent. As a result, clock switching may not be performed in time after a clock fault occurs, reducing reliability. In eCPRI MUX scenarios, the clock quality level of the multimode RF module depends on the clock of the RAT providing optical fiber connections. If the clock of the RAT providing optical fiber connections is faulty, the AAU clock is abnormal and clock switching cannot be performed. In eCPRI MUX scenarios, the clock unlock function is not supported. For details about eCPRI MUX scenarios, see *eCPRI*.

In the dual-fiber and single-mode AAU networking scenarios, the impacts of eCPRI clock unlock and clock lock on wireless services vary with the situations of inter-RAT interference (upgrade, reset, restart, or fault), link switchover, and clock quality deterioration. [Table 4-5](#) lists the differences.

Table 4-5 Impacts of eCPRI clock unlock and clock lock on wireless services

eCPRI Clock Unlock	eCPRI Clock Lock
The modes do not affect each other. Especially in the single-mode AAU networking scenario, wireless services are affected only by the devices of the local mode.	Services on the single-mode AAU on the BBU involved in clock lock are greatly affected. When the main control board of the major standard for clock lock is upgraded and reset, services of the single-mode AAU of the minor standard for clock lock are affected. After clock lock is disabled, services are intermittently interrupted.
A link fault causes a clock switchover. The impact of the switchover is the same as that in the clock lock scenario.	A link fault causes a clock switchover. The clock switchover requires re-lock. The impact lasts about 30 seconds.
1. In the dual-fiber networking scenario, the impact of clock quality deterioration is the same as that in the eCPRI clock lock scenario. 2. In the single-mode AAU networking scenario, only services of the local mode are affected.	1. In the dual-fiber networking scenario, the impact of clock quality deterioration is the same as that in the eCPRI clock unlock scenario. 2. In the single-mode AAU networking scenario, clock quality deterioration has a great impact on services. When the mode is the major standard for clock lock and the clock quality deteriorates to a certain extent, interference to neighboring devices occurs.

As shown in the preceding table, the difference between eCPRI clock unlock and eCPRI clock lock is that eCPRI clock lock has a great impact on wireless services in single-mode AAU scenarios.

NOTICE

After the clock unlock switch is turned on, the clock changes from the lock mode to the unlock mode. In this case, the eCPRI clock on the AAU side may be switched. For example, the AAU obtains clock signals from the LTE mode instead of from the NR mode. Clock switching causes clock phase difference and service loss.

If a clock switchover occurs on the clock link traced by the AAU due to a clock or port exception, services will be interrupted when the clock is re-traced and locked.

4.2 Network Analysis

4.2.1 Benefits

When system clock lock optimization is enabled, the major standard for clock lock can be preferentially determined based on the clock quality to ensure the optimal clock performance.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations (macro base stations)
UMTS	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

GSM must use the GTMUc as the main control board.

The UMTS/LTE/NR/co-MPT side must use the UMPT as the main control board.

RF Modules

This function does not depend on RF modules.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 When to Use

If a separate-MPT multimode base station uses the dual-star or CPRI MUX networking and is enabled with the GU@5 MHz or GSM and LTE Spectrum Concurrency feature, the clock of each standard must strictly follow frequency synchronization and phase synchronization. That is, the system clocks among all RATs must be strictly synchronized. To achieve this, the system clocks among all RATs in a multimode base station must be mutually locked.

4.4.2 Precautions

When inter-RAT clock lock is applied, the following are true:

- In BBU interconnection scenarios, the major standard for clock lock must be located in the primary BBU.
- If the GBTS is deployed first and only the hardware installation of the peer RAT is complete, GSM serves as the major standard for clock lock.

- The type of the main control board where the major standard for clock lock is located determines the holdover duration in which the system can work properly. For example, in GL clock lock scenarios, if the GTMU serves as the main control board for the major standard for clock lock and the reference source is lost, the holdover duration of LTE TDD services depends on the GTMU holdover capability, which is 8 hours. For details about the holdover duration, see *Synchronization*.

4.4.3 Data Configuration

4.4.3.1 Data Preparation

System Clock Lock Optimization

Run the **SET BTSOTHPARA** command on the GBTS to enable system clock lock optimization.

Parameter Name	Parameter ID	Setting Notes
Clock Mutual Lock Optimization Switch	BTSOTHPARA <i>.ClkMutualLockOptSw</i>	Set this parameter to ON . This parameter controls whether to enable the clock lock optimization function. If this parameter is set to OFF , GSM in a separate-MPT base station must be the major standard for clock lock. If this parameter is set to ON , GSM in a separate-MPT base station can be either the major or minor standard for clock lock. This function can be enabled when GSM in a separate-MPT base station uses a GTMUc as the main control board.

Forcible Major Standard for Clock Lock

Run the **SET CLKSYNCMODE** command on the base station side to set the forcible major standard for clock lock.

Parameter Name	Parameter ID	Setting Notes
Forcibly Major Standard for Clock Lock	TASM.FIXED MS (5G gNodeB, LTE eNodeB)	This parameter specifies the forcible major standard for clock lock. If this parameter is set to AUTO , the major standard for clock lock is automatically selected. If this parameter is set to a value other than AUTO , the setting takes effect only when the following conditions are met: (1) In co-BBU LNR separate-MPT scenarios, the clock synchronization modes of the two RATs are the same. (2) Neither of the two RATs is configured with the CGPS or RGPS clock source.

Clock Unlock in eCPRI Scenarios

Run the **SET CLKSYNCMODE** command on the LTE and NR sides to turn on the eCPRI clock unlock switch.

Parameter Name	Parameter ID	Setting Notes
eCPRI Clock Unlock Switch	TASM.ECPRID EMULLOCKS W (5G gNodeB, LTE eNodeB)	This parameter controls whether to enable the eCPRI clock unlock function. When this switch is turned on, the eCPRI clock unlock function is enabled for the local base station. When this switch is turned off, the eCPRI clock unlock function is disabled for the local base station.

4.4.3.2 Using MML Commands

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

System Clock Lock Optimization

```
SET BTSOTHPARA: CLKMUTUALLOCKOPTSW=ON;
```

Forcible Major Standard for Clock Lock

```
//Setting the forcible major standard for clock lock (LTE used as an example)  
SET CLKSYNCMODE: FIXEDMS=LTE;
```

Clock Unlock in eCPRI Scenarios

```
SET CLKSYNCMODE: CLKSYNCMODE=TIME, ECPRIDEMULLOCKSW=ON;
```

4.4.4 Activation Verification

System Clock Lock Optimization

- Step 1** On the GBTS side, run the **DSP BTSBRD** command with the **INFOTYPE** parameter as well as the GTMU cabinet, subrack, and slot numbers specified to query the major standard for clock lock.
- Step 2** On the UMTS/LTE/NR/co-MPT side, run the **DSP CLKMUTUALLOCK** command to query the major standard for clock lock.

If...	Then...
The queried major standard for clock lock is consistent with that described in 4.1.3.1 System Clock Lock of a Separate-MPT Base Station Including GBTS	The configuration is successful.
The queried major standard for clock lock is inconsistent with that described in 4.1.3.1 System Clock Lock of a Separate-MPT Base Station Including GBTS	The configuration fails.

- Step 3** Check whether any alarm is generated.

If...	Then...
ALM-28020 Site Configuration Error is generated with the value of Configuration Item being Clock Mutual Lock Optimization Configuration	The configuration fails.

----End

Forcible Major Standard for Clock Lock

After the forcible major standard for clock lock is configured, run the **DSP CLKMUTUALLOCK** command on the LTE or NR side. If the query result shows that the major standard for clock lock is the configured standard, the configuration is successful.

Clock Unlock in eCPRI Scenarios

- Step 1** Run the **LST CLKSYNCMODE** command on the LTE and NR sides to query the clock unlock switch. If the switch is turned on on both sides, eCPRI clock unlock is enabled.
- Step 2** On the LTE/NR/co-MPT side, run the **DSP CLKMUTUALLOCK** command to query the major standard for clock lock.

If...	Then...
The major standard for clock lock does not exist, and NULL is displayed.	The configuration is successful.
The major standard for clock lock exists and is not displayed as NULL .	The configuration does not take effect. There may be constraints described in 4.1.4 Clock Unlock in eCPRI Scenarios .

- Step 3** On the LTE/NR/co-MPT side, run the **DSP CLKSTAT** command with the cabinet number, subrack number, and slot number of the AAU specified to query the AAU clock status.

If...	Then...
Current Clock Source State is Normal , and PLL Status is Locked .	The configuration is successful.
AAU Clock Port is 0 or 1 .	Run the DSP RRU command with Cabinet No. , Subrack No. , and Slot No. specified to query the port number of the RRU of the local mode. If the port number is the same as the value of AAU Clock Port for the AAU, the AAU uses the clock of the local mode.

----End

4.4.5 Network Monitoring

- On the GBTS:

If...	Then...
ALM-28020 Site Configuration Error is generated with the value of Configuration Item being Clock Mutual Lock Optimization Configuration	The GTMU does not support the system clock lock optimization function.

If this alarm is reported, check the GTMU hardware type. If it is not GTMUc, replace the GTMU in use with a GTMUc board.

- On the base station:

If...	Then...
ALM-26245 Configuration Data Inconsistency is reported	The configuration of the forcible major standard for clock lock does not take effect.
ALM-26276 Inter-System Site-Level Configuration Conflict is reported	If the forcible major standard for clock lock is configured on both the LTE and NR main control boards and the configured major standards are inconsistent, the earlier configuration takes effect on one base station, and this alarm is reported on the other base station.

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- *Common Transmission* for SingleRAN
- *Synchronization* for GSM BSS, WCDMA RAN, eRAN, and 5G RAN
- *eXn Self-Management* for 5G RAN